EEE 431
ELECTROMAGNETIC FIELD THEORY
LECTURE NOTES

PRESENTATION

By

Dr. Aliyu Buba Abdullahi

MSc. Ph.D. AFHEA FHEA MIET MIEEE

Electrical & Electronics, School of Engineering,
Federal Polytechnic Mubi, PMB 35 Adamawa State

CONTENTS

- Aims & Objectives
- Learning Objectives
- Teaching/Learning Methods
- Assessment Criteria
- Topic area to cover (I – IV)
- Problem Solutions

AIMS & OBJECTIVES

MODULE AIM:

To understand the principle and application of:

1. ELECTROSTATICS
2. STATIC MAGNETIC FIELDS
3. TIME VARRYIN ELECTROMAGNETIC FIELDS
4. PLANE WAVES

LEARNING OBJECTIVES

At the end of the class, you will be able to:

- **Know the, principles and applications of electrostatics, and solve problems.**
- **Know the, principles and applications of static magnetic fields, and solve problems.**
- **Know the principles and applications of time-varying electromagnetic fields, and solve problems**
- **Know the principles and applications of plane waves, and solve problems.**

TEACHING/LEARNING METHODS

- LECTURE SESSIONS
- TUTORIAL & WORKSHOPS
- INDEPENDENT STUDY

LEARNING OUTCOMES

1. Be able to produce routine solutions to basic electrostatics, static magnetic fields, time varying electromagnetic fields, plane waves.
2. Understand the laws and apply them to solve problems related to these electro-magnetic fields.

2. Be able to demonstrate ability to meet the training needs of individuals within the communications industry.

ASSESSMENT CRITERIA

- **ASSESSMENT CRITERIA**
  1. **Course Works (Assignments).**
  2. **Time constrained Classroom tests.**
  3. **End of Semester Examination**
     (by the end of the session)

LECTURE SESSIONS

- ❑ Introduction:
  - ✓ Course Brief And Recommended Textbook
  - ✓ Introduction To (Scalar And) Vector Analysis
- ❑ Understanding Principles & Applications of
  1. Electrostatics
  2. Static Magnetic Fields
  3. Time Varying Electromagnetic Fields
  4. Plane Waves

COURSE BRIEF

Electromagnetic field theory: study of charges at rest/motion

- ✓ **study of electrical engineering**

- **Required for the understanding**…

  Electrical, Electromechanical Systems

Electromagnetic field Deals Directly with the electric and magnetic field vectors

- ✓ **Problems:**

Usually involve three space variables along with the time variable…

- Vector analysis as the convenient tool to comprehend

COURSE BRIEF

Electromagnetic field theory: consists of static electric fields, static magnetic fields, time-varying fields & it's applications.

Applications: **<u>Electrostatic fields</u>**

- **Deflection of a charged particle.**
- **Sorting of minerals e.g. ore separation.**
- **Electrostatic generator and electrostatic voltmeter**

Applications: **<u>Static magnetic fields</u>**

- **static magnetic fields are in dc machines.**
- **magnetic deflection, magnetic separator, cyclotron, hall effect sensors, magneto hydrodynamic generator etc.**

COURSE BRIEF

- <u>**Recommended Textbooks:**</u>

[1]. W. H. Hayt (Jr), J. A. Buck, "Engineering Electromagnetics", TMH

[2]. K. E. Lonngren, S.V. Savor, "Fundamentals of Electromagnetics with Matlab", PHI

[3]. E.C. Jordan, K.G. Balmain, "Electromagnetic Waves & Radiating System", PHI.

[4]. M. N. Sadiku, "Elements of Electromagnetics", Oxford University Press.

[5] D. Tougaw "Applied Electromagnetic Field Theory" Valparaiso University Press

INTRODUCTION

Vector Analysis

Electromagnetic field theory deal with quantities (either scalar or vectors)

Scalars → characterized by magnitude only

Vectors→ both magnitude and direction

A vector $\vec{A}$ → $\vec{A} = \hat{a}\, A$

$A = |\vec{A}|$ **= magnitude**

$\hat{a} = \frac{\vec{A}}{|A|}$ **= unit vector (i.e., unit magnitude and same direction**

INTRODUCTION

Vector Analysis

Two vectors $\vec{A}$ and $\vec{B}$ can be added together to give another vector $\vec{C}$

$$\vec{C} = \vec{A} + \vec{B}$$

Scaling of a vector $\rightarrow$ $\vec{C} = \alpha\vec{B}$ where $\vec{C}$ is scaled version of vector $\vec{B}$ and α is a scalar

INTRODUCTION

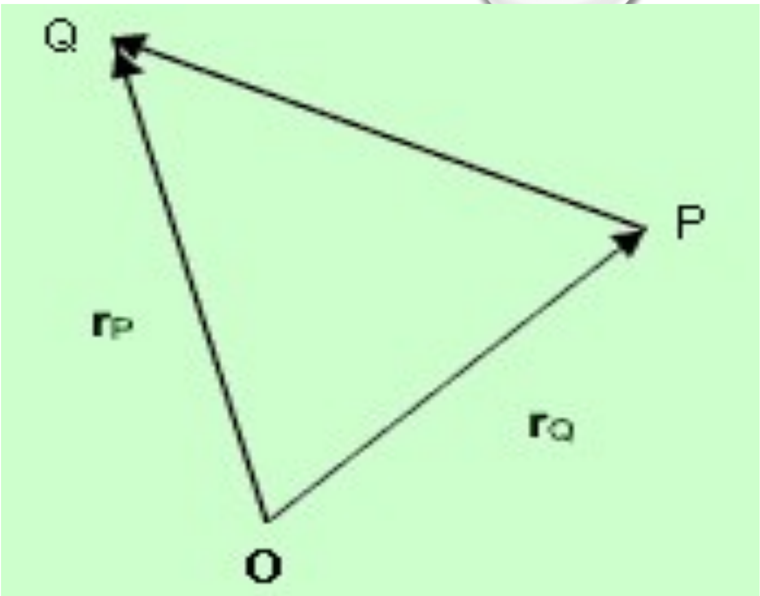
Vector Analysis

Some important laws of vector algebra…

$$\vec{A}+\vec{B}=\vec{B}+\vec{A}$$ ………*(Commutative Law)*

$$\vec{A}+\left(\vec{B}+\vec{C}\right)=\left(\vec{A}+\vec{B}\right)+\vec{C}$$ ………*(Associative Law)*

$$\alpha\left(\vec{A}+\vec{B}\right)=\alpha\vec{A}+\alpha\vec{B}$$ ……….*(Distributive Law)*



INTRODUCTION

Vector Analysis

Position of a vector $\vec{r}_Q$ of a point P is the directed distance from the origin (O) to P, i.e.,

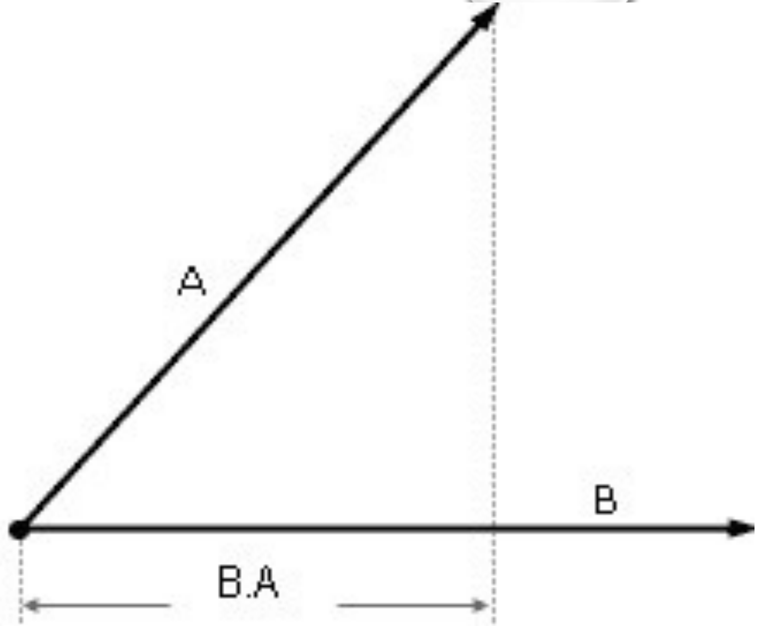
$\vec{r}_Q = \overrightarrow{OP}$ **If** $\vec{r}_Q = OP$ $\quad \vec{r}_P = OQ$ are the position vectors

of the points P and Q then the distance vector

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \vec{r}_P - \vec{r}_Q$$

INTRODUCTION

Product of Vector



Two vectors $\vec{A}$ and $\vec{B}$ are multiplied, the result is either a scalar or a vector depending how the two vectors were multiplied

Scalar product (or dot product) $\vec{A} \cdot \vec{B} = |A||B|cos\theta_{AB}$

Vector product (or cross product) $\vec{A} \times \vec{B} = |A||B|\sin\theta_{AB} \cdot \vec{n}$

$\vec{n}$ is unit vector perpendicular to $\vec{A}$ and $\vec{B}$

INTRODUCTION

Product of Vector

More detailed of the vector productors in the lecture notes…

Covering:

- **Cumulative dot product**
- **Scalar/vector triple products**
- **Coordinate system**
- **Del Operator**
- **Gradient of scalar**
- **Divergence theorem**
- **Stokes Theorem**
- **Numerical Examples**

MODULE CONTENT

To understand the principle and application of:

1. ELECTROSTATICS
   - ✓ Coulomb's LAW
   - ✓ Point charges and related field
   - ✓ Gauss’s LAW etc.
2. STATIC MAGNETIC FIELDS
3. TIME VARRYIN ELECTROMAGNETIC FIELDS
4. PLANE WAVES

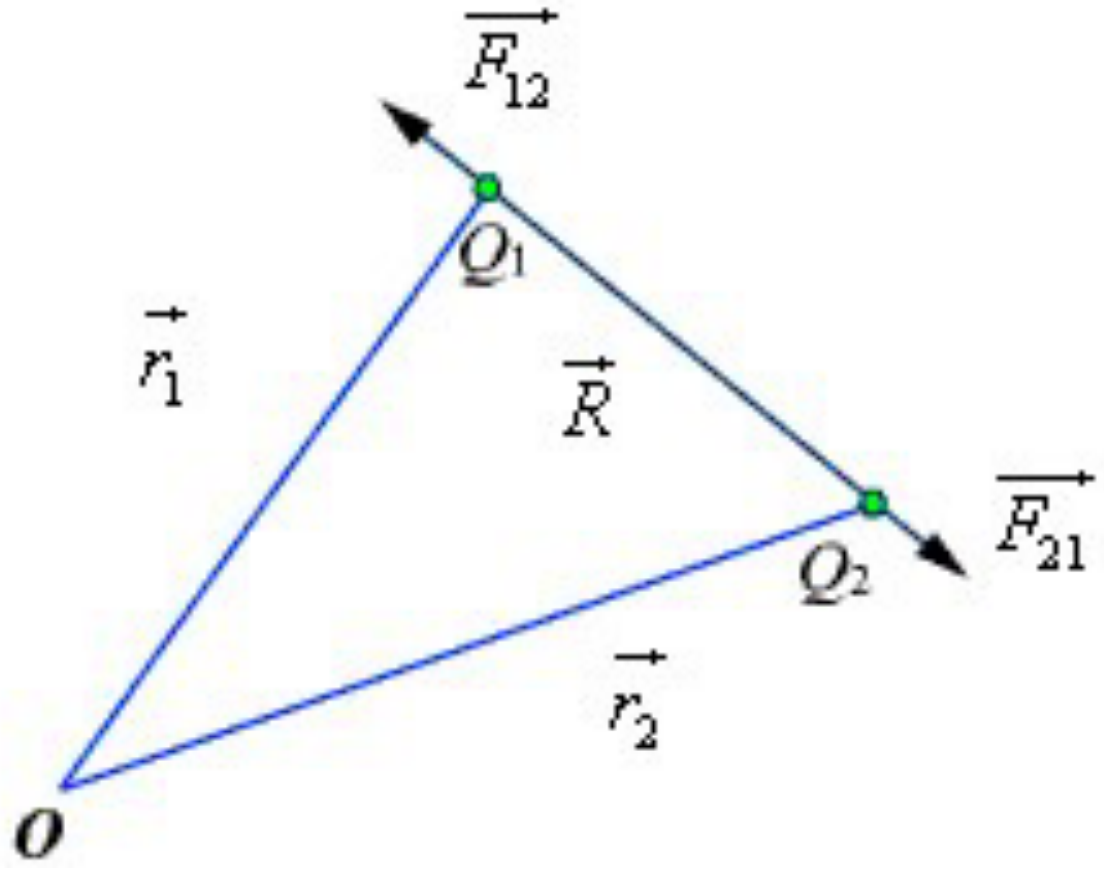
TOPIC-1 AREAS:

ELECTROSTATIC:

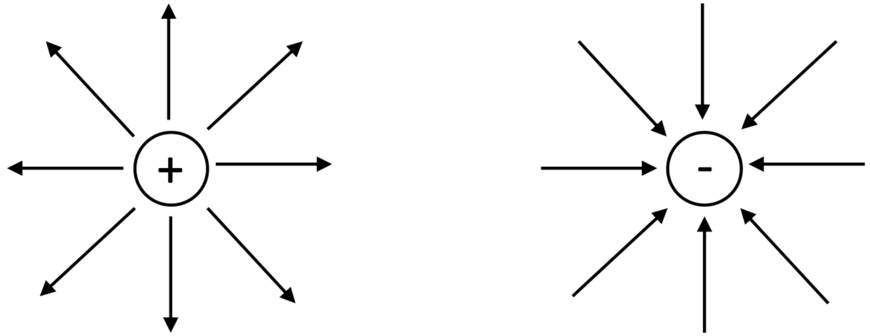
- Coulomb’s LAW
- Force on point Charge
- Intensity of Electric Field
- Flux density
- Gauss’s LAW
- Applications

ELECTROSTATICS

COULOMB's law:

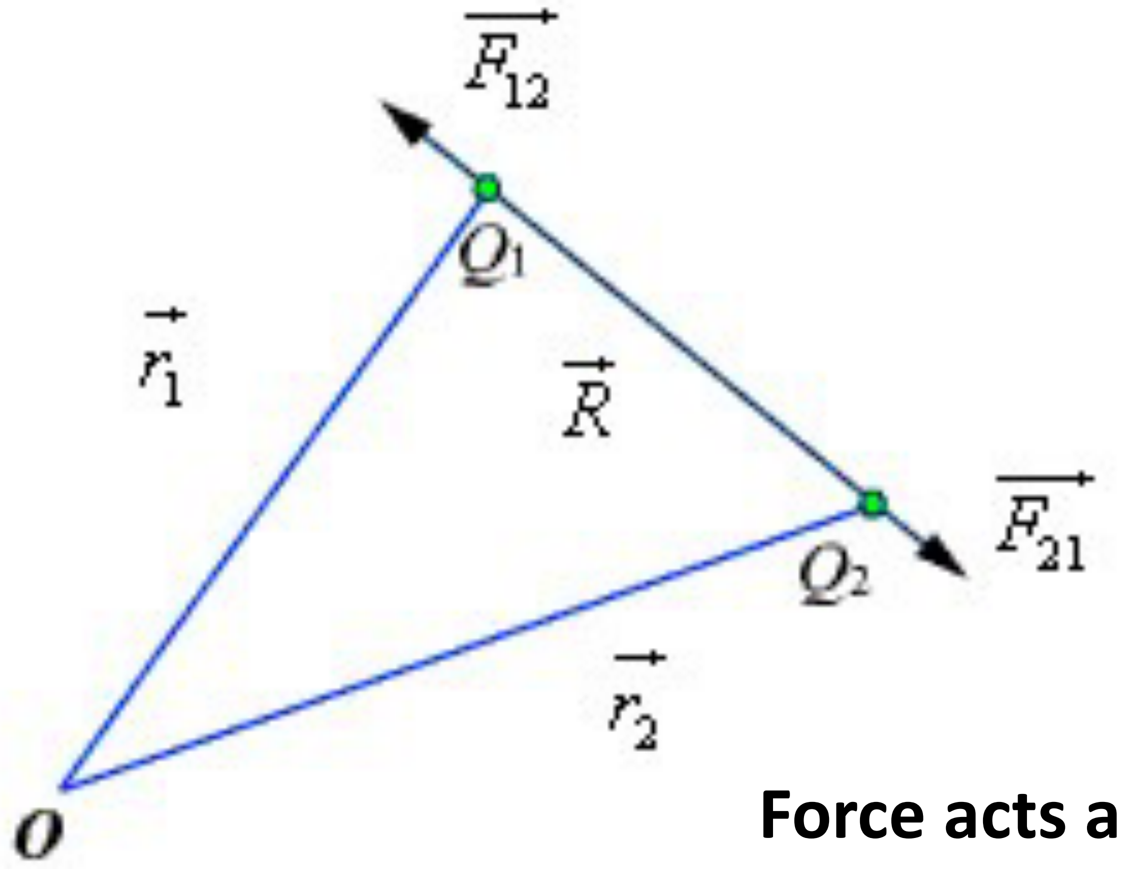


- **law expresses the force between two point charges Q_1, and Q_2.**
- **Charges polarity matters: if these charges are likes/unlike, they attract or repel each other.**



ELECTROSTATICS

COULOMB's law:

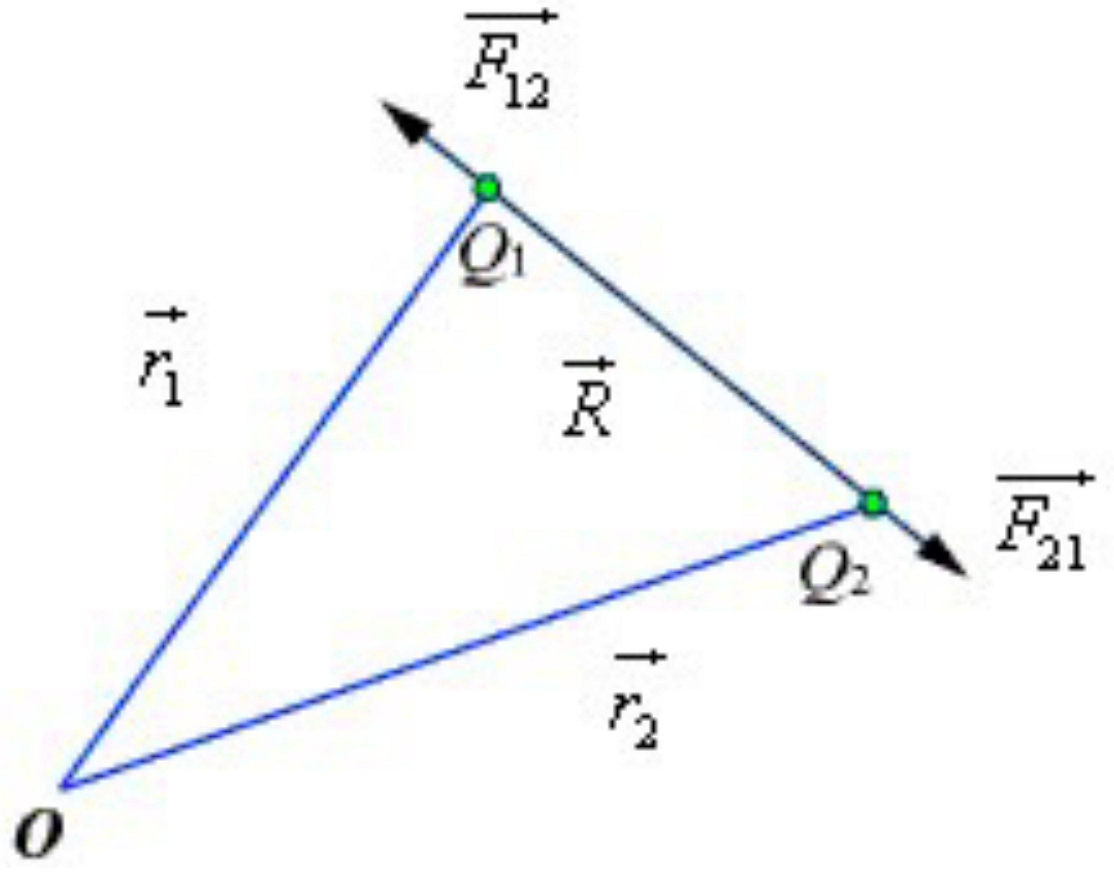


…the force between two-point charges Q_1,and Q_2separated in vacuum or free space by a distance R which is large compared to their size… is directly proportional to the product of their charges and inversely proportional to the square of the distance between them.

Force acts along the line joining the two charges.

ELECTROSTATICS

COULOMB's law:



Mathematically:

$$F = \frac{kQ_1Q_2}{R^2}$$

Where:

Q_1 and Q_2 = charge (C)

F = force (N)

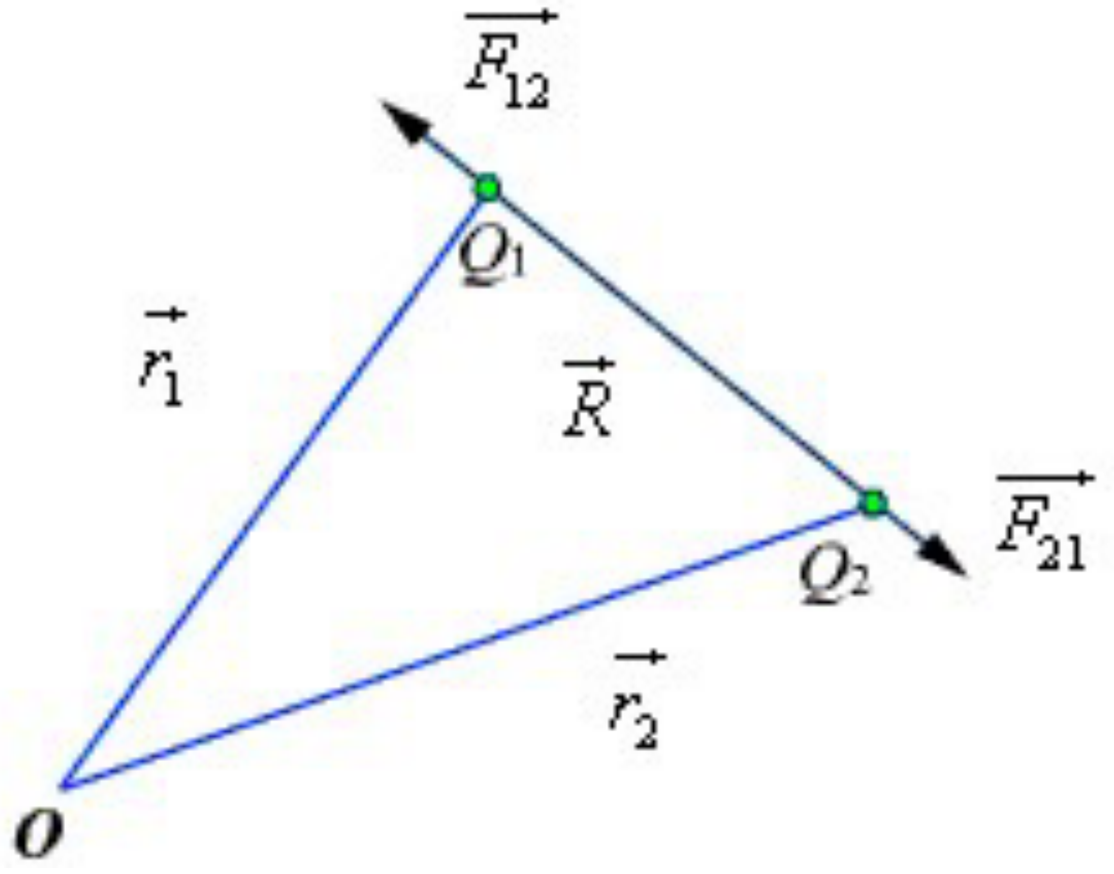
k = constant proportionality $k = \frac{1}{4\pi\varepsilon_o}$

ε_o = permittivity of free space

8.85×10^{-12}= $\frac{1}{36\pi}\times10^{-9}$F/m

ELECTROSTATICS

COULOMB's law:



Unit vector:

- let the position vectors of the point charges $Q_1$and Q_2 are given by $\vec{r_1}$ **and** $\vec{r_2}$

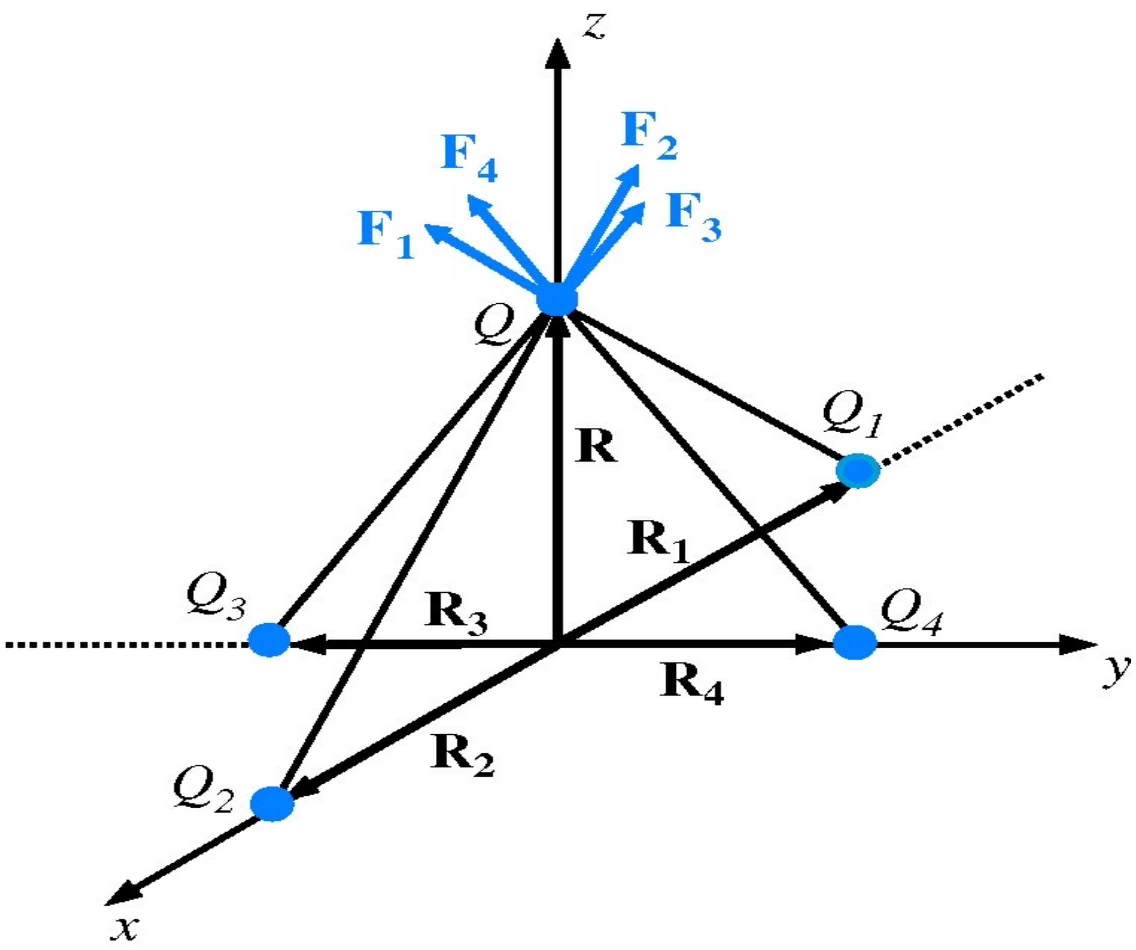
- Let $\overrightarrow{F_{12}}$ represent the force on Q_1 due to charge Q_2

Then we can define a unit vectors as:

$$\widetilde{a_{12}} = \frac{\left(\vec{r_2} - \vec{r_1}\right)}{R} \quad \text{and} \quad \widetilde{a_{21}} = \frac{\left(\vec{r_1} - \vec{r_2}\right)}{R}$$

ELECTROSTATICS

COULOMB's law:



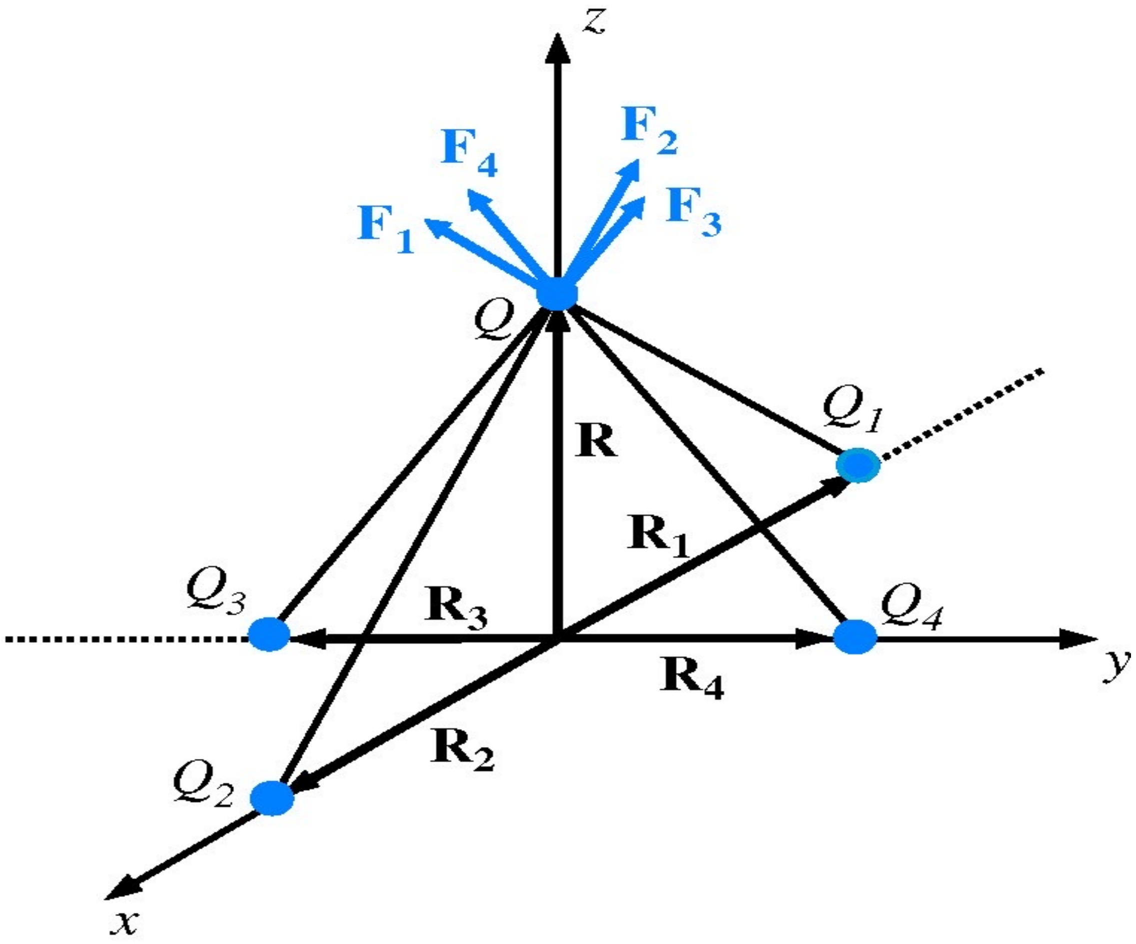
Example:

Four charges of 10 μC each are located in free space at points with Cartesian coordinates $(-3,0,0)$, $(3,0,0)$, $(0,-3,0)$, and $(0,3,0)$.

Find the force on a 20-μC charge located at $(0,0,4)$. All distances are in meters.

ELECTROSTATICS

COULOMB's law:

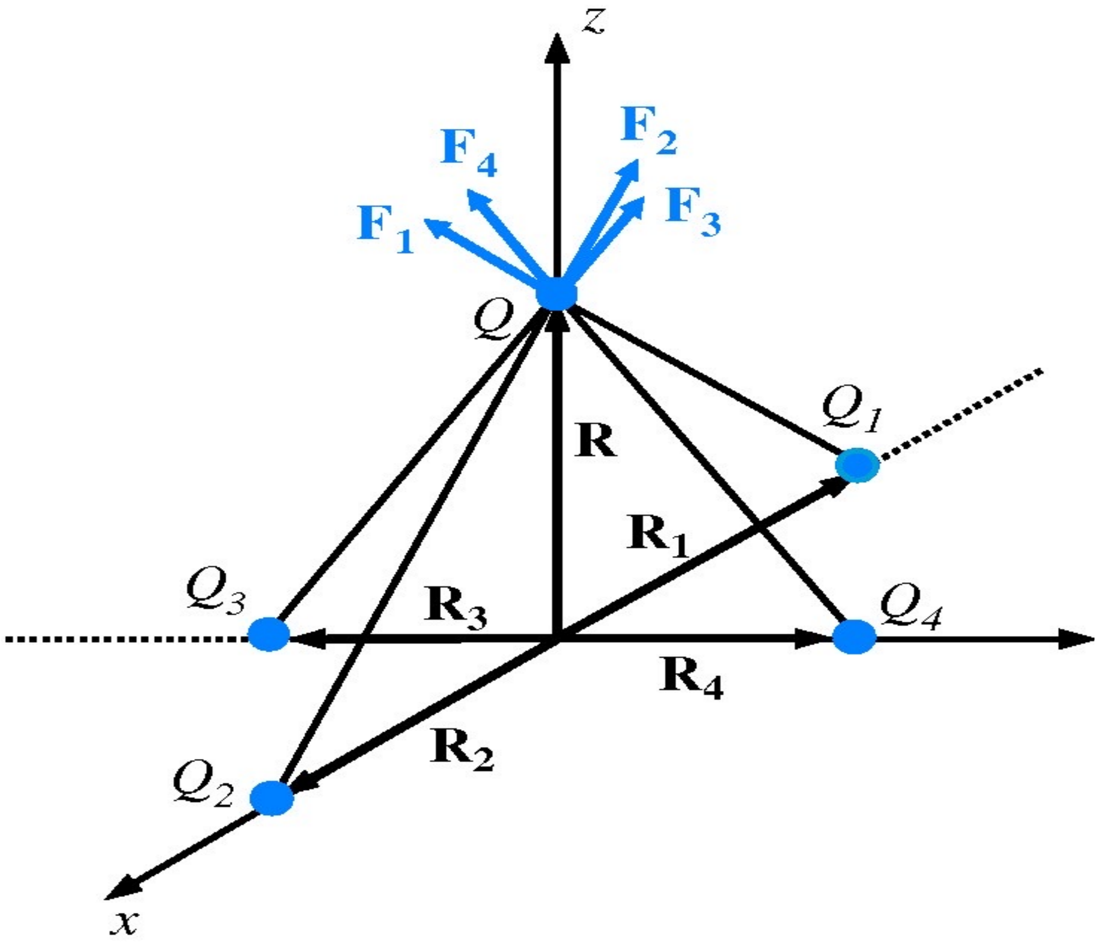


Solution:

$$R_1 = -x\ 3$$
$$R_2 = x\hat{}\ 3$$
$$R_3 = -y\hat{}\ 3$$
$$R_4 = y\hat{}\ 3$$
$$R = z\hat{}4$$

ELECTROSTATICS

COULOMB's law:



Solution:

$$\mathbf{F}_1 = \frac{QQ_1}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_1}{|\mathbf{R}-\mathbf{R}_1|^3} = \frac{QQ_1}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4+\hat{\mathbf{x}}3}{125} = \frac{QQ_1}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4+\hat{\mathbf{x}}3)$$

$$\mathbf{F}_2 = \frac{QQ_2}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_2}{|\mathbf{R}-\mathbf{R}_2|^3} = \frac{QQ_2}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4-\hat{\mathbf{x}}3}{125} = \frac{QQ_2}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4-\hat{\mathbf{x}}3)$$

$$\mathbf{F}_3 = \frac{QQ_3}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_3}{|\mathbf{R}-\mathbf{R}_3|^3} = \frac{QQ_3}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4+\hat{\mathbf{y}}3}{125} = \frac{QQ_3}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4+\hat{\mathbf{y}}3)$$

$$\mathbf{F}_4 = \frac{QQ_4}{4\pi\varepsilon_0}\frac{\mathbf{R}-\mathbf{R}_4}{|\mathbf{R}-\mathbf{R}_4|^3} = \frac{QQ_4}{4\pi\varepsilon_0}\frac{\hat{\mathbf{z}}4-\hat{\mathbf{y}}3}{125} = \frac{QQ_4}{500\pi\varepsilon_0}(\hat{\mathbf{z}}4-\hat{\mathbf{y}}3)$$

$$\begin{aligned}\mathbf{F} &= \mathbf{F}_1+\mathbf{F}_2+\mathbf{F}_3+\mathbf{F}_4 \\ &= \frac{200\times 10^{-12}}{500\pi\varepsilon_0}(\hat{\mathbf{z}}16) = \hat{\mathbf{z}}\frac{32\times 10^{-12}}{5\pi\times 8.85\times 10^{-12}} = \hat{\mathbf{z}}0.23 \quad \text{(N)}.\end{aligned}$$

ELECTROSTATICS

INTENSITY OF ELECTRIC FIELD:

Electric field intensity E (or the electric field strength) at a point is defined as the force per unit charge.

Mathematically: $\vec{E} = \lim_{Q \to 0} \frac{\vec{F}}{Q}$ or, $\vec{E} = \frac{\vec{F}}{Q}$

E at a point r **(observation point)** due a point charge *Q* located at $\vec{r}\,'$ (source point)

$$\vec{E} = \frac{Q(\vec{r} - \vec{r}\,')}{4\pi\varepsilon_0 \left|\vec{r} - \vec{r}\,'\right|^3}$$

ELECTROSTATICS

INTENSITY OF ELECTRIC FIELD:

For a collection of *N* point charges $Q_1, Q_2, \quad ...Q_N$ located at $\vec{r}_1, \vec{r}_2, ...\vec{r}_N$

E at a point r **(observation point)** due to these charge $Q_1, Q_2, \quad ...Q_N$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{Q_k(\vec{r}-\vec{r}_i)}{\left|\vec{r}-\vec{r}_i\right|^3}$$

ELECTROSTATICS

INTENSITY OF ELECTRIC FIELD:

For an elementary charge $dQ = \rho(\vec{r'})dv'$ considering this as point charge, we can write the field expression as:

$$d\vec{E} = \frac{dQ(\vec{r} - \vec{r'})}{4\pi\varepsilon_0 \left|\vec{r} - \vec{r'}\right|^3} = \frac{\rho(\vec{r'})dv'(\vec{r} - \vec{r'})}{4\pi\varepsilon_0 \left|\vec{r} - \vec{r'}\right|^3}$$

If integrated over the source region, we get the electric field at the point *P* due to this distribution of charges.

$$\overrightarrow{E(r)} = \int_V \frac{\rho(\vec{r'})(\vec{r} - \vec{r'})}{4\pi\varepsilon_0 \left|\vec{r} - \vec{r'}\right|^3} dv'$$

ELECTROSTATICS

INTENSITY OF ELECTRIC FIELD:

Example: A square plate in the x–y plane is situated in the space defined by $-3\ \text{m} \le x \le 3\ \text{m}$ and $-3\ \text{m} \le y \le 3\ \text{m}$. **Find the total charge on the plate if the surface charge density is given by** $\rho_s = 4y^2\ (\mu\text{C/m}^2)$.

Solution:

$$\rho_s = 4y^2$$

$$Q = \int_S \rho_s\, ds$$

$$= \int_{-3}^{3}\int_{-3}^{3} 4y^2\, dx\, dy$$

$$= \left.\left.\frac{4y^3x}{3}\right|_{-3}^{3}\right|_{-3}^{3} = 432\ \mu\text{C} = 0.432 \quad (\text{mC}).$$

ELECTROSTATICS

ELECTRIC FLUX DENSITY:

E depends on the material media in which the field is being considered… thus, flux density vector is defined to be independent of the material media.

$$\vec{D} = \varepsilon \vec{E}$$

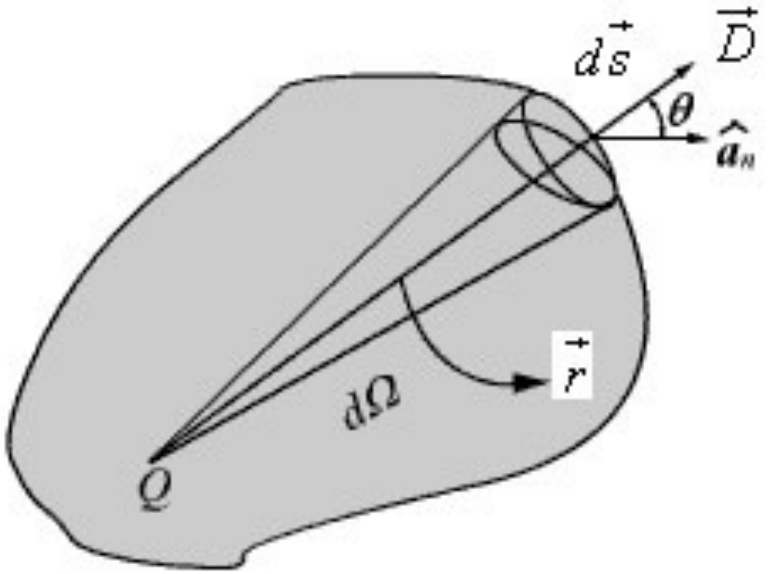
We define the electric flux ψ as

$$\psi = \int_S \vec{D} . d\vec{s}$$

ELECTROSTATICS

Gauss's LAW:

Gauss law states that…the total electric flux through a closed surface is equal to the total charge enclosed by the surface.

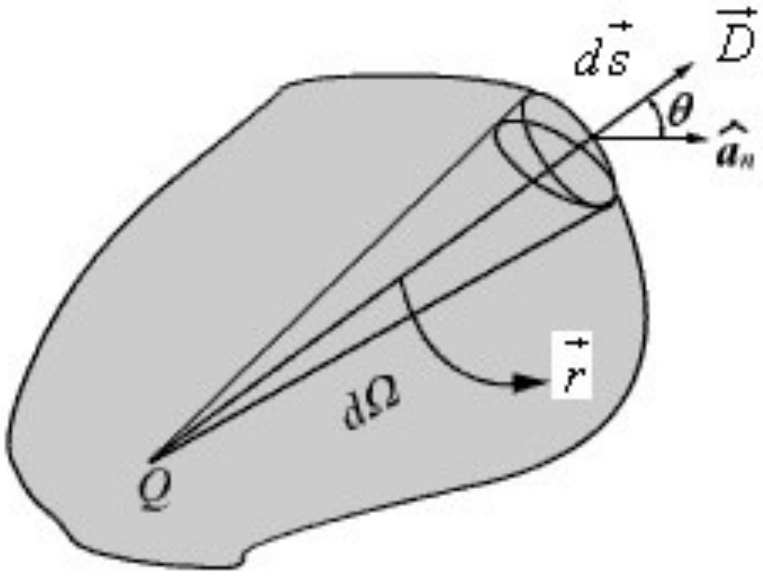


$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$$

ELECTROSTATICS

Gauss's LAW:

Gauss law states that…the total electric flux through a closed surface is equal to the total charge enclosed by the surface**.**



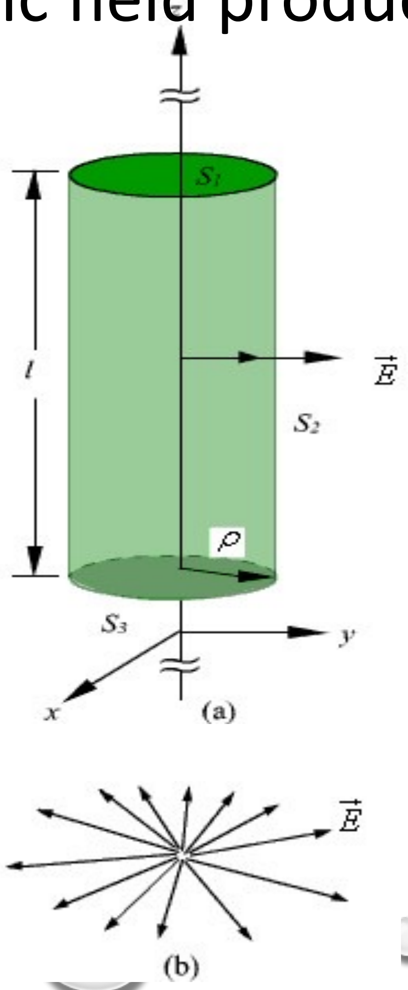
Applications.

particularly useful in computing $\vec{E}$ or $\vec{D}$ where the charge distribution has some symmetry

ELECTROSTATICS

Gauss's LAW:

Let's consider the problem of determining the electric field produced by an infinite line charge of density r_LC/m.



This line charge positioned along the *z*-axis as shown in Fig. (a).

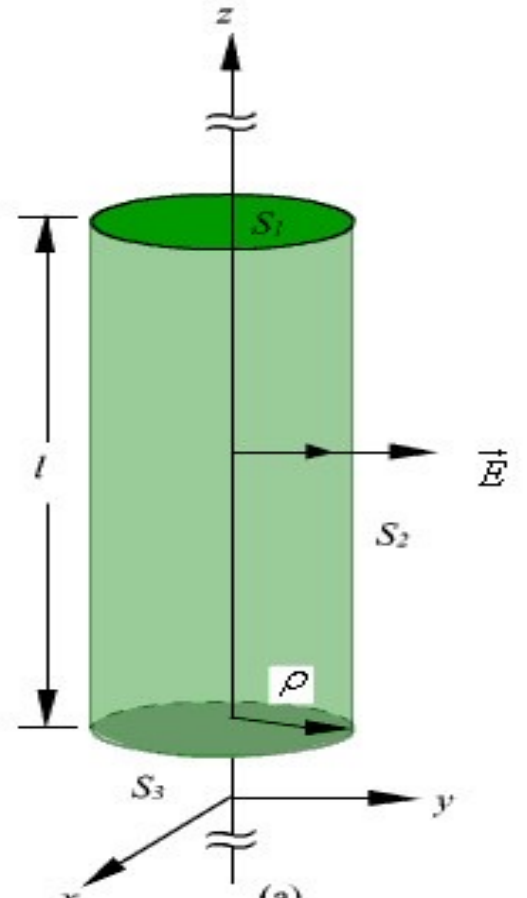
Since the line charge is assumed to be infinitely long, the electric field will be of the form as shown in Fig. 1.2(b).

ELECTROSTATICS

Gauss's LAW:

If we consider a close cylindrical surface as shown in Fig. (a), using Gauss's theorem we can write,.

$$\rho_L l = Q = \oint_S \varepsilon_0 \vec{E}.d\vec{s} = \int_{S_1} \varepsilon_0 \vec{E}.d\vec{s} + \int_{S_2} \varepsilon_0 \vec{E}.d\vec{s} + \int_{S_3} \varepsilon_0 \vec{E}.d\vec{s}$$



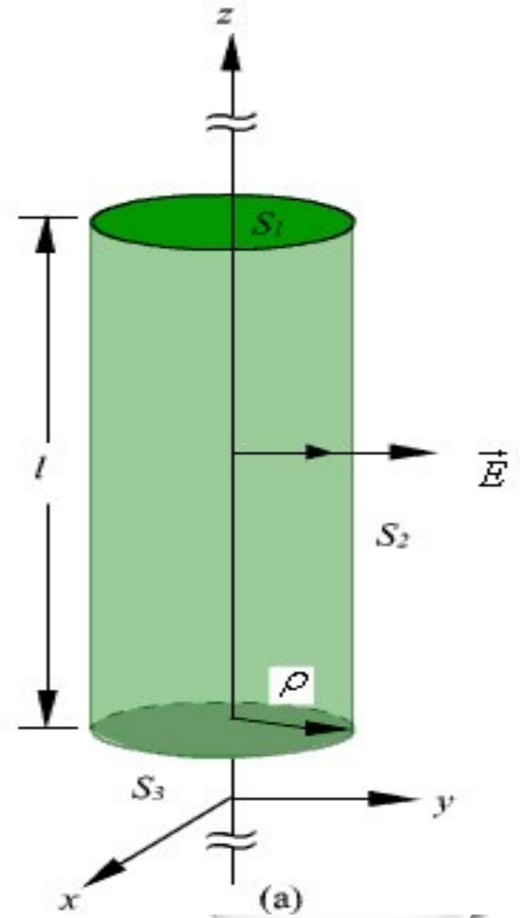
(a)

ELECTROSTATICS

Gauss's LAW:

$$\rho_L l = Q = \oint_S \varepsilon_0 \vec{E}.d\vec{s} = \int_{S_1} \varepsilon_0 \vec{E}.d\vec{s} + \int_{S_2} \varepsilon_0 \vec{E}.d\vec{s} + \int_{S_3} \varepsilon_0 \vec{E}.d\vec{s}$$

Because the unit normal vector to areas S_1 and S_3 are perpendicular to the electric field, the surface integrals for the top and bottom surfaces evaluates to zero.



Hence, we can write, $\rho_L l = \varepsilon_0 E.2\pi\rho l$

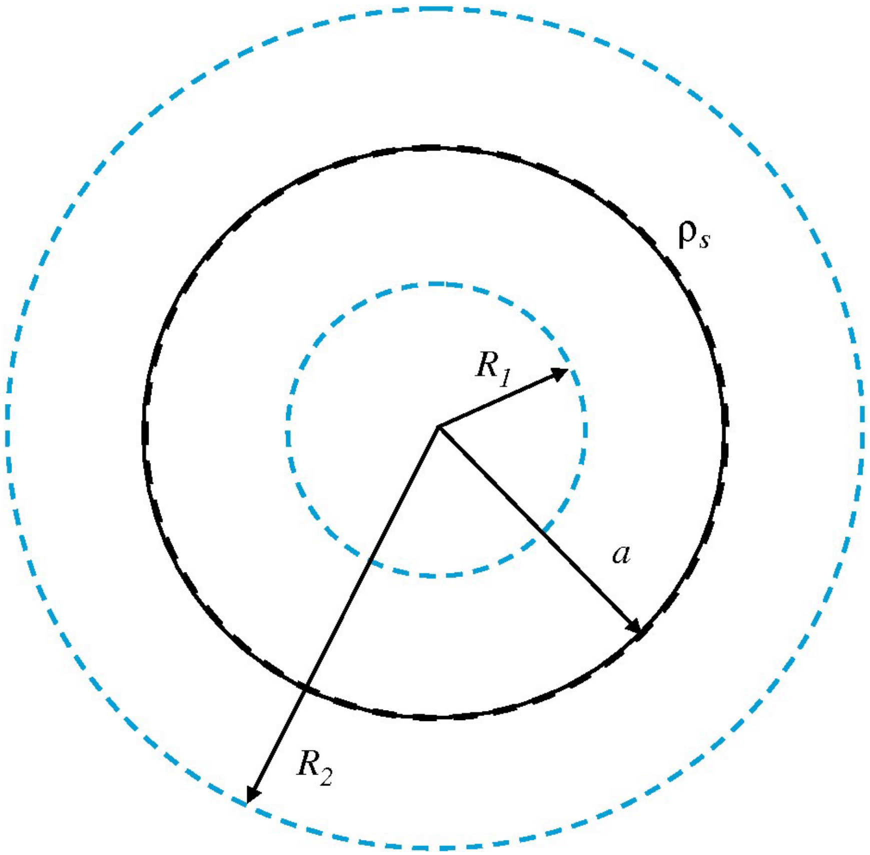
$$\vec{E} = \frac{\rho_L}{2\pi\varepsilon_0 \rho}\hat{a}_\rho$$

ELECTROSTATICS

Gauss's LAW:

Example: A thin spherical shell of radius *a* carries uniform surface charge density ρs. **Use Gauss's law to determine E.**

Solution:



$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$$

Symmetry suggest D is radial in direction

$$\mathbf{D} = \hat{\mathbf{R}} D_R$$

$$d\mathbf{s} = \hat{\mathbf{R}}\, ds$$

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \oint_S D_R\, ds = D_R(4\pi R^2) = Q$$

$$D_R = \frac{Q}{4\pi R^2}$$

ELECTROSTATICS

Gauss's LAW:

Example: A thin spherical shell of radius *a* carries uniform surface charge density ρs. **Use Gauss's law to determine $\boldsymbol{E}$.**

Solution:

- For a Gaussian surface of radius $R_1 < a$, no charge is enclosed. Hence, $Q = 0$, in which case $E = 0$.
- For a Gaussian surface of radius $R_2 > a$,

$$Q = \rho_s(4\pi a^2)$$

and

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon} = \frac{\hat{\mathbf{R}}}{\varepsilon} D_r = \frac{\hat{\mathbf{R}} Q}{4\pi\varepsilon R_2^2} = \hat{\mathbf{R}} \frac{4\pi\rho_s a^2}{4\pi\varepsilon R_2^2} = \hat{\mathbf{R}} \frac{\rho_s a^2}{\varepsilon R_2^2}\ .$$

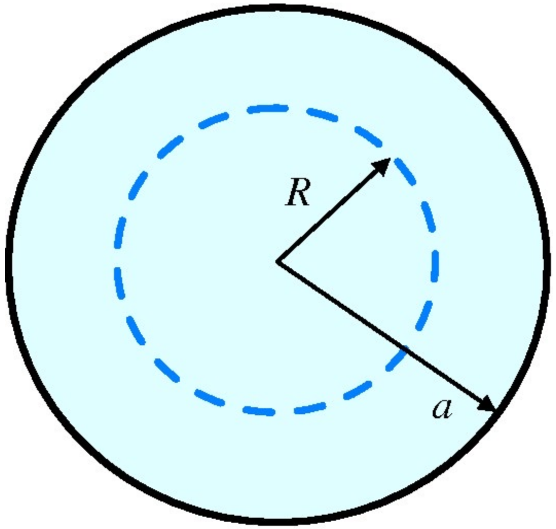
ELECTROSTATICS

Gauss's LAW:

Example: A spherical volume of radius a contains a uniform volume charge density ρv. **Use Gauss's law to determine D for (a) R ≤ a and (b) R ≥ a.**

Solution: For $R \leq a$,

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \oint_S D_r \, ds = D_r (4\pi R^2)$$



$R < a$

Q within a sphere of radius R is

$$Q = \frac{4}{3}\pi R^3 \rho_v$$

Hence,

$$4\pi R^2 D_R = \frac{4}{3}\pi R^3 \rho_v$$

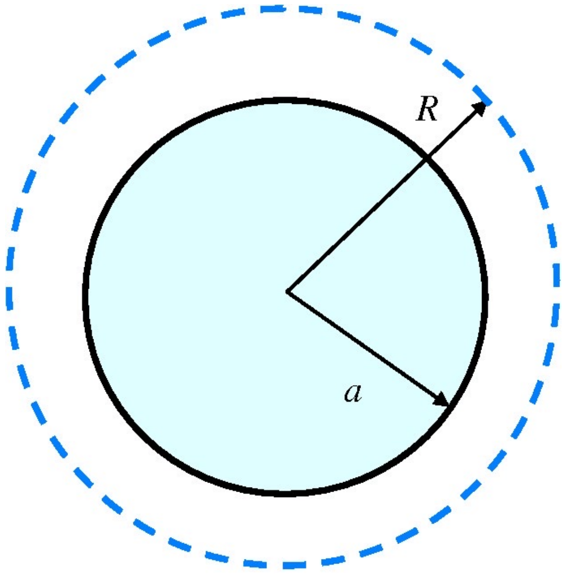
$$D_r = \frac{\rho_v R}{3}, \qquad \mathbf{D} = \hat{\mathbf{R}} D_r = \hat{\mathbf{R}} \frac{\rho_v R}{3}, \qquad R \leq a.$$

ELECTROSTATICS

Gauss's LAW:

Example: A spherical volume of radius a contains a uniform volume charge density ρv. **Use Gauss's law to determine D for (a) R ≤ a and (b) R ≥ a.**

Solution: For $R \geq a$, total charge in sphere is



$$Q = \frac{4}{3}\pi a^3 \rho_v,$$

$$4\pi R^2 D_R = \frac{4}{3}\pi a^3 \rho_v,$$

$$\mathbf{D} = \hat{\mathbf{R}} D_r = \hat{\mathbf{R}} \frac{\rho_v a^3}{3R^2}, \qquad R \geq a.$$

ELECTROSTATICS

Examples: READING ASSIGNMENT 1 **& 2:**

Some examples in the lecture notes and summary presentation.

Solution HINTS:

MODULE CONTENT

To understand the principle and application of:

1. ~~ELECTROSTATICS~~
2. STATIC MAGNETIC FIELDS (MAGNETOSTATICS)
   - ✓ Biot-Savart's Law
   - ✓ Ampere's law:
   - ✓ Derivation of:
     i. Divergence's Theorem
     ii. Stoke's Theorem
   - ✓ Magnetic Flux Density
3. TIME VARRYIN ELECTROMAGNETIC FIELDS
4. PLANE WAVES

TOPIC-2 AREAS:

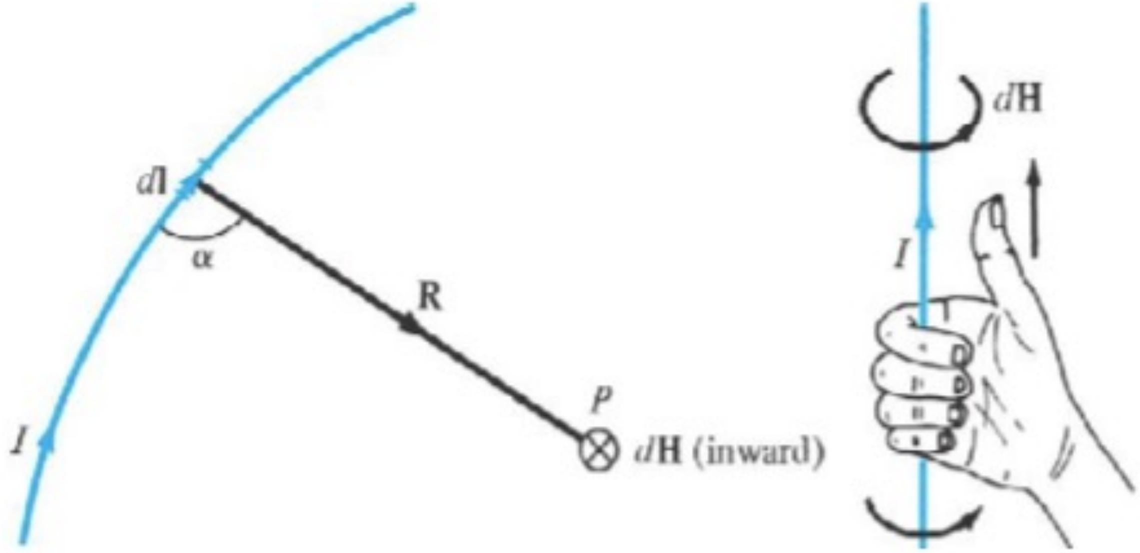
MAGNETOSTATIC:

- Biot-Savart’s Law
- Ampere’s law
- Derivation of:
    1. Divergence’s Theorem
    2. Stoke’s Theorem
- Amperes law from Stoke’s theorem
- Application of Amperes LAW
- Magnetic Flux Density etc.

STATIC MAGNETIC FIELD

Biot-Savart's LAW:

This state that… the differential magnetic field intensity, dH, produced at a point P, by the differential current element, Idl, is proportional to the product Idl and the sine of the angle between the element and the line joining P to the element and is inversely proportional to the square of the distance, R, between P and the element .

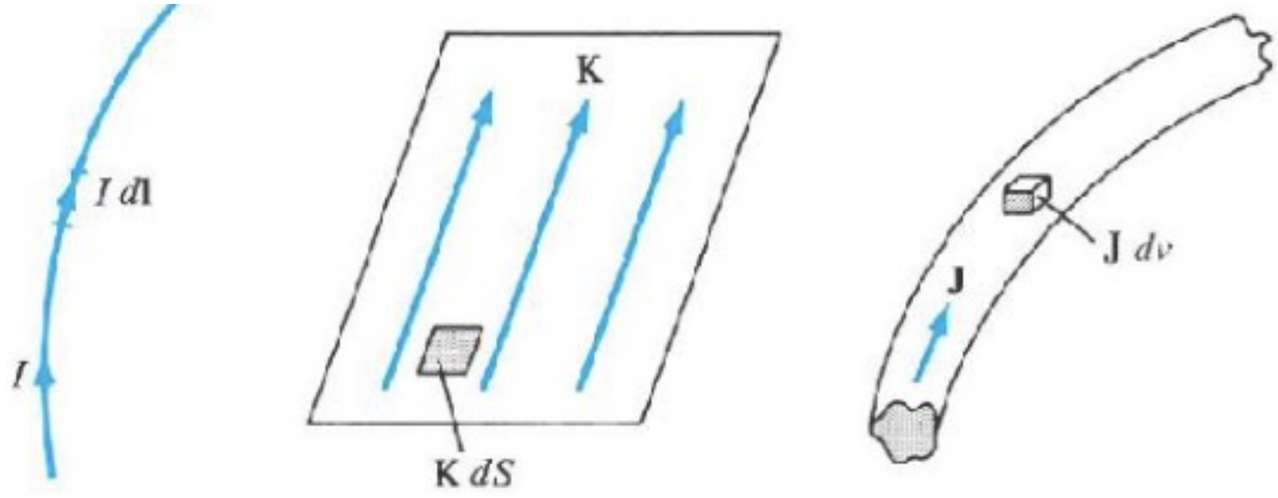


$$d\vec{H} = \frac{Id\vec{l} \times \hat{a}_R}{4\pi R^2} = \frac{Id\vec{l} \times \bar{R}}{4\pi R^3} = \frac{Idl \sin\alpha}{4\pi R^2}$$

STATIC MAGNETIC FIELD

Biot-Savart's LAW:

Considering different current distributions (as shown above) we can rewrite expression for field intensity as below …



$$I d\vec{l} = \vec{K} dS = \vec{J} dv$$

$$\vec{H} = \int_L \frac{I d\vec{l} \times \hat{a}_R}{4\pi R^2}$$

$$\vec{H} = \int_S \frac{\vec{K} dS \times \hat{a}_R}{4\pi R^2}$$

$$\vec{H} = \int_v \frac{\vec{J} dv \times \hat{a}_R}{4\pi R^2}$$

STATIC MAGNETIC FIELD

H FIELD From a straight (Filament):

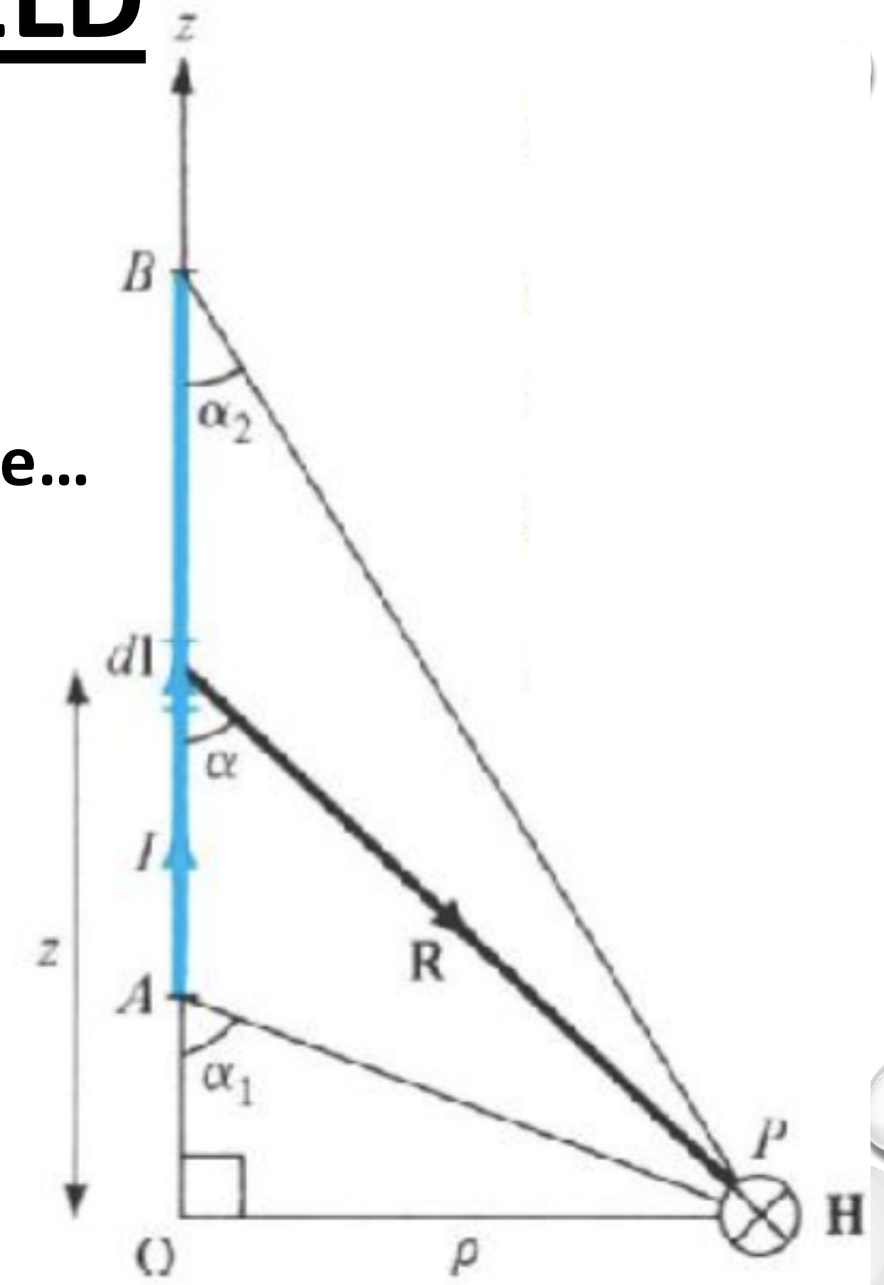
The H field is determined for a strait filament of current in a manner very similar to that of the electric field determined from a line charge…

$$\vec{H} = \int_L \frac{I d\vec{l} \times \vec{R}}{4\pi R^3}$$

$$d\vec{l} = dz\hat{a}_z$$

$$\vec{R} = \rho\hat{a}_\rho - z\hat{a}_z$$

$$d\vec{l} \times \vec{R} = \rho dz\hat{a}_\phi$$



STATIC MAGNETIC FIELD

H FIELD From a straight (Filament):

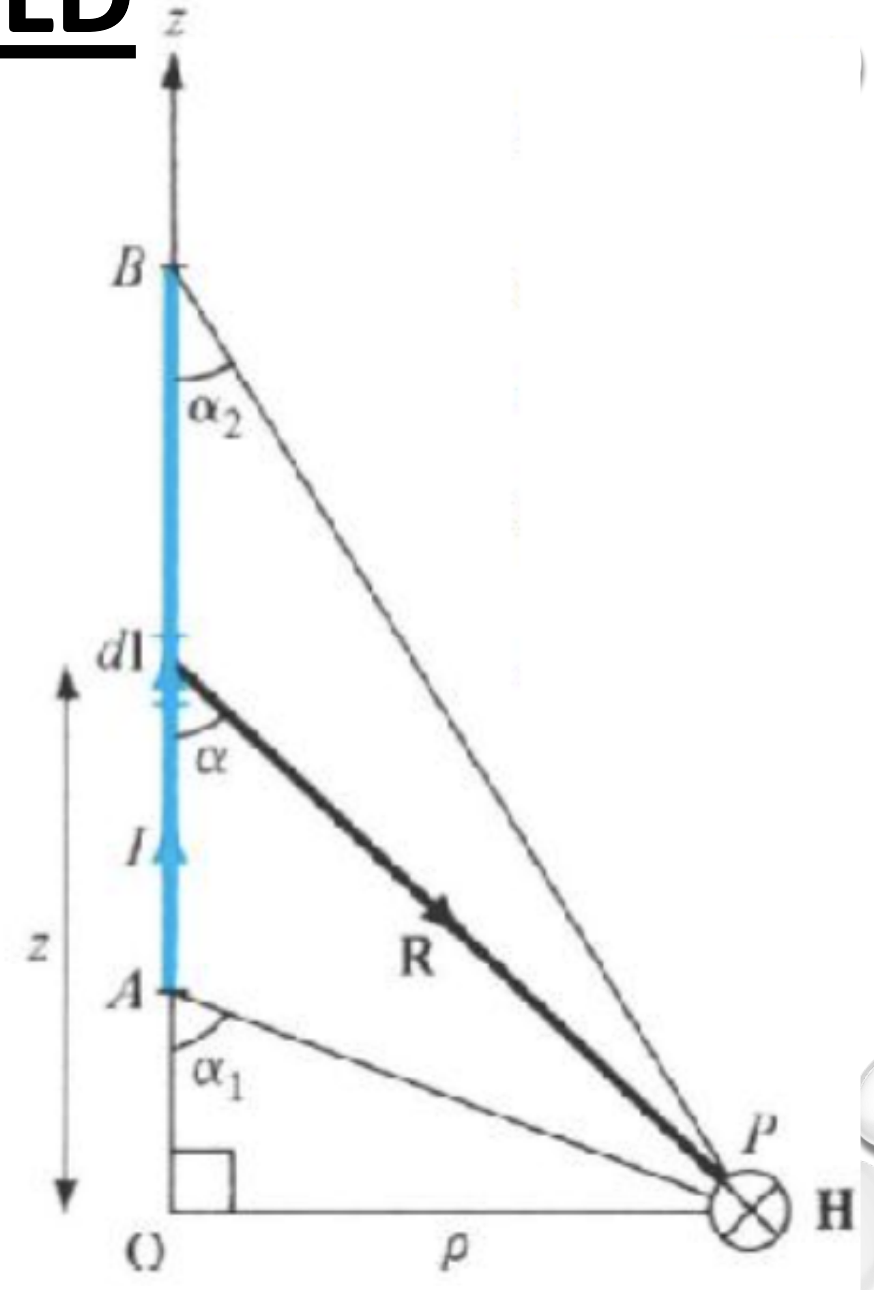
$$\vec{H} = \int_L \frac{I\rho dz\hat{a}_\phi}{4\pi\left(\rho^2 + z^2\right)^{3/2}}$$

Now:

$$z = \rho \cot\alpha$$

$$dz = -\rho\left(\csc^2\alpha\right)d\alpha$$

$$\vec{H} = \int_L -\frac{I\rho^2\left(\csc^2\alpha\right)d\alpha\hat{a}_\phi}{4\pi(\rho\csc\alpha)^3}$$



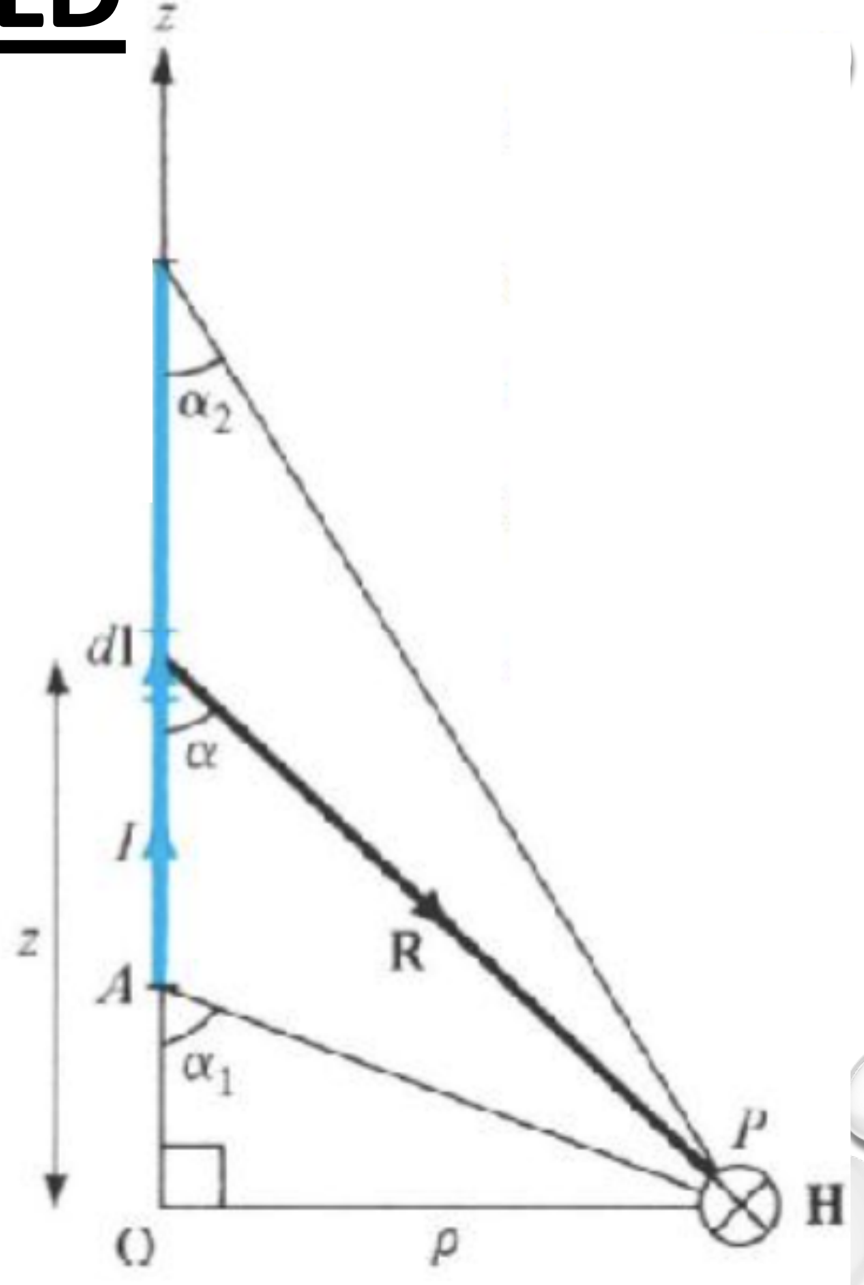
STATIC MAGNETIC FIELD

H FIELD From a straight (Filament):

$$\vec{H} = \int_L -\frac{I\rho^2(\csc^2\alpha)d\alpha\hat{a}_\phi}{4\pi(\rho\csc\alpha)^3}$$

$$\vec{H} = -\frac{I}{4\pi\rho}\int_{\alpha_1}^{\alpha_2}(\sin\alpha)d\alpha\hat{a}_\phi$$

$$\vec{H} = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\hat{a}_\phi$$

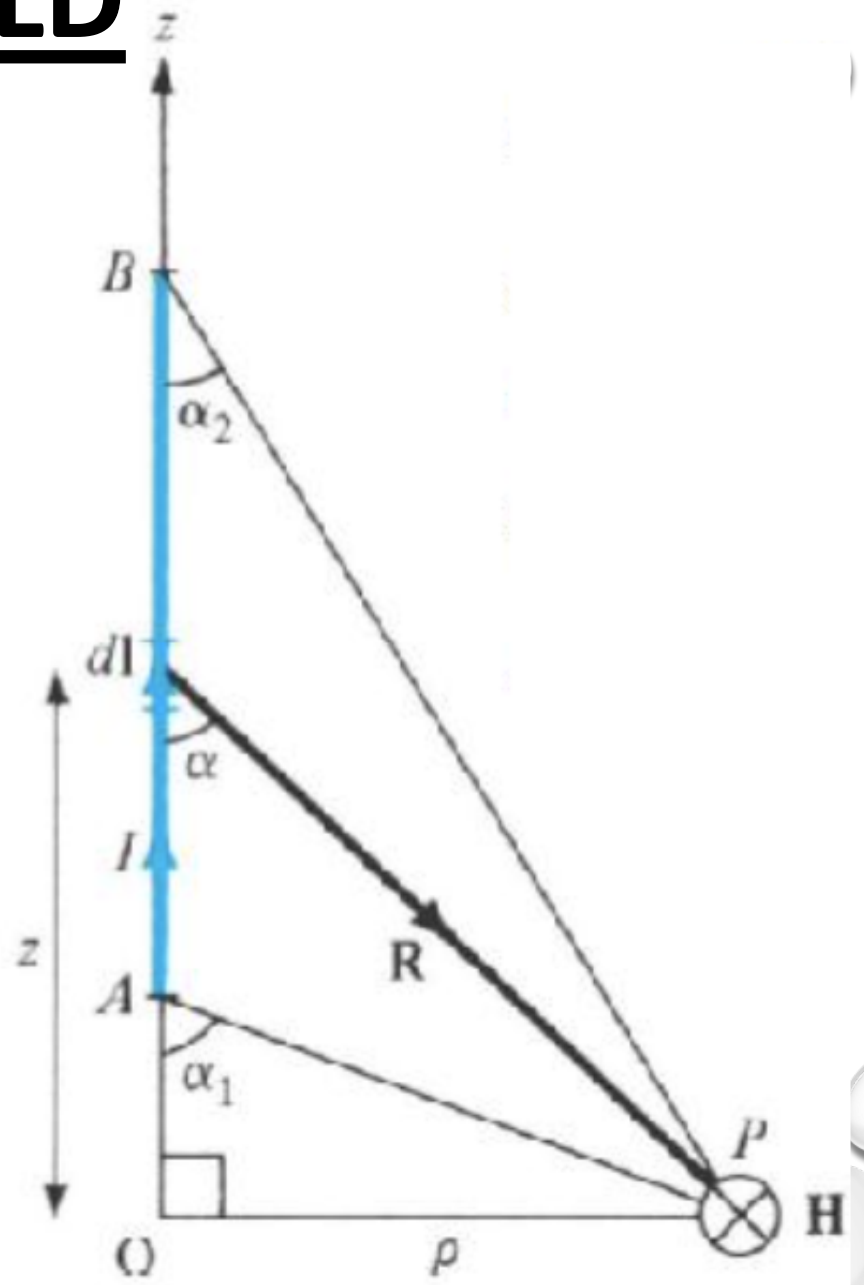


STATIC MAGNETIC FIELD

H FIELD From a straight (Filament):

$$\vec{H} = \frac{I}{4\pi\rho}\hat{a}_\phi$$ Line from z = 0 to ∞

$$\vec{H} = \frac{I}{2\pi\rho}\hat{a}_\phi$$ Line from z = -∞ to ∞



Using Biot-savart's law we can find the expression for field intensity due to different current carrying conductor configurations.

STATIC MAGNETIC FIELD

Ampere's LAW:

This state that… The line integral of H around a closed path is the same as the net current, I_{enc} enclosed by the path.

$$\oint \vec{H} \cdot d\vec{l} = I_{enc}$$

- **This law is easily used to determine H when the current distribution is symmetrical.**
- **It ALWAYS holds, even if the current distribution is NOT symmetrical, however the equation is typically used for symmetric cases.**
- **Ampere's law is a special case of the Biot-Savart law.**

STATIC MAGNETIC FIELD

Derivation of Divergence's and Stoke's Theorem: :

Recalled from the VECTOR introduction class the details on the following:

A divergence of a Vector A, at any given point P is the outward flux per unit volume as volume shrinks about P.

$$div\vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \to 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

STATIC MAGNETIC FIELD

Divergence's Theorem:

The divergence theorem states that…

the total outward flux of a vector field, A, through the closed surface, S, is the same as the volume integral of the divergence of A.

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

$$div\vec{A} = \nabla \cdot \vec{A} = \lim_{\Delta v \to 0} \frac{\oint_s \vec{A} \cdot d\vec{S}}{\Delta v}$$

$$\int_v \nabla \cdot \vec{A} dv = \oint_s \vec{A} \cdot d\vec{S}$$

STATIC MAGNETIC FIELD

Stoke’s Theorem:

Stokes theorem states that…

the circulation of a vector field A, around a closed path, L is equal to the surface integral of the curl of A over the open surface S bounded by L.

This theorem has been proven to hold as long as A and the curl of A are continuous along the closed surface S of a closed path L.

STATIC MAGNETIC FIELD

Stoke's Theorem:

This theorem has been proven to hold as long as A and the curl of A are continuous along the closed surface S of a closed path L.

$$\vec{A} = A_1\hat{a}_1 + A_2\hat{a}_2 + A_3\hat{a}_3$$

$$curl\vec{A} = \nabla \times \vec{A} = \left(\lim_{\Delta S \to 0} \frac{\oint_L \vec{A}\cdot d\vec{l}}{\Delta S} \right)_{\max} \hat{a}_n$$

$$\oint_L \vec{A}\cdot d\vec{l} = \oint_s \left(\nabla \times \vec{A}\right)\cdot d\vec{S}$$

STATIC MAGNETIC FIELD

Applying Stokes's theorem on Ampere's law:

Applying Stokes's theorem on Ampere's law, we have,

$$I_{enc} = \oint_L \vec{H} \cdot d\vec{l} = \int_S \left(\nabla \times \vec{H}\right) \cdot d\vec{S}$$

We can also write using current density:

$$I_{enc} = \int_S \vec{J} \cdot d\vec{S}$$

So, from the above two equations:

$$\nabla \times \vec{H} = \vec{J}$$

STATIC MAGNETIC FIELD

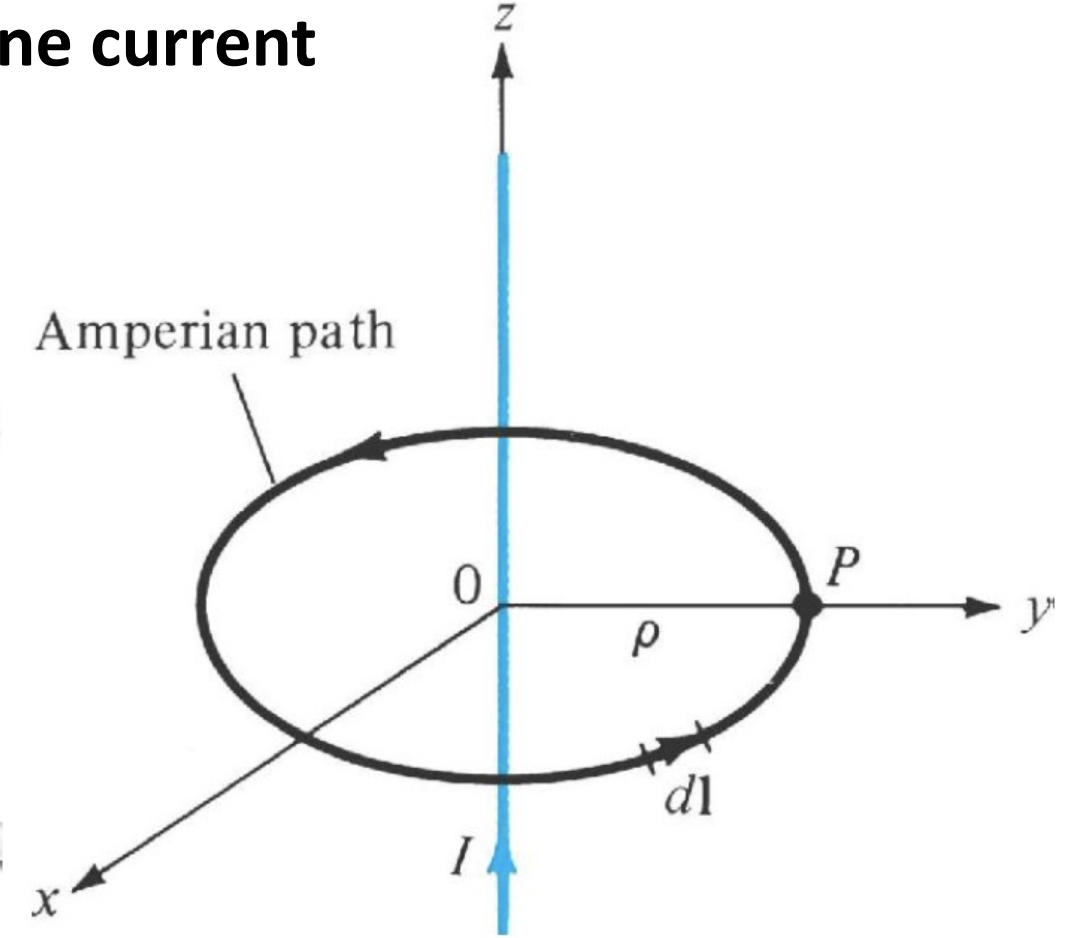
Application of Ampere's Circuital law:

A simple application of Ampere's law can be used to easily derive the magnetic field intensity from an infinite line current

$$I_{enc} = \oint \vec{H} \cdot d\vec{l}$$

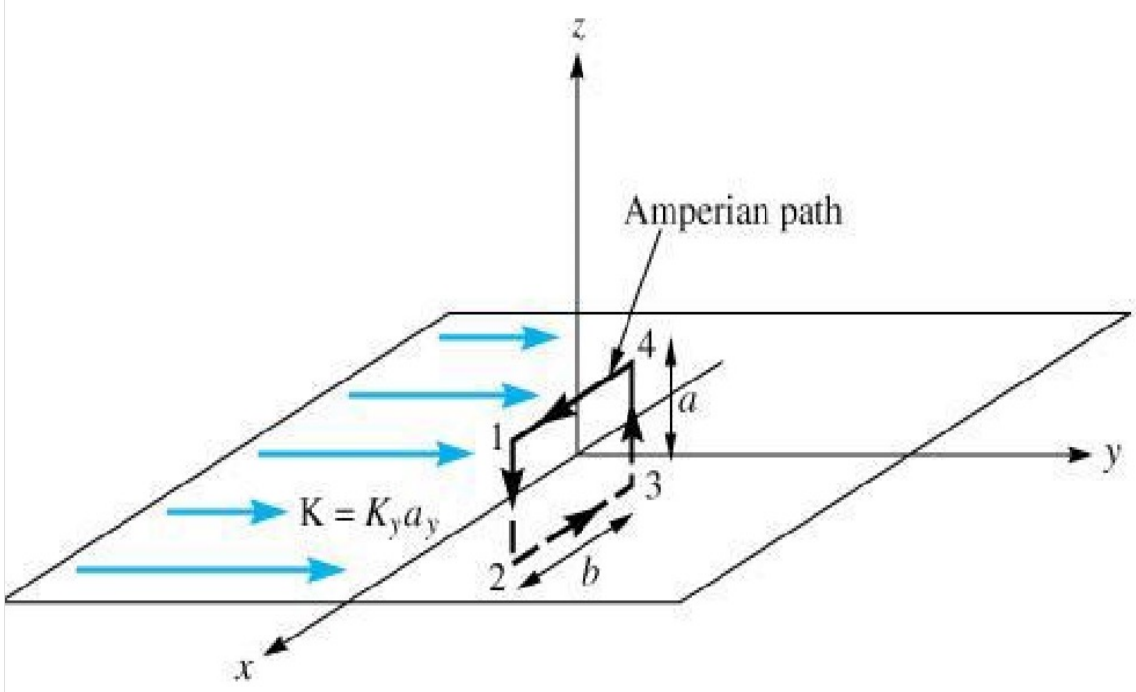
$$I = \int H_{\phi}\hat{a}_{\phi} \cdot \rho d\phi \hat{a}_{\phi} = H_{\phi}\int \rho d\phi = H_{\phi} \cdot 2\pi\rho$$

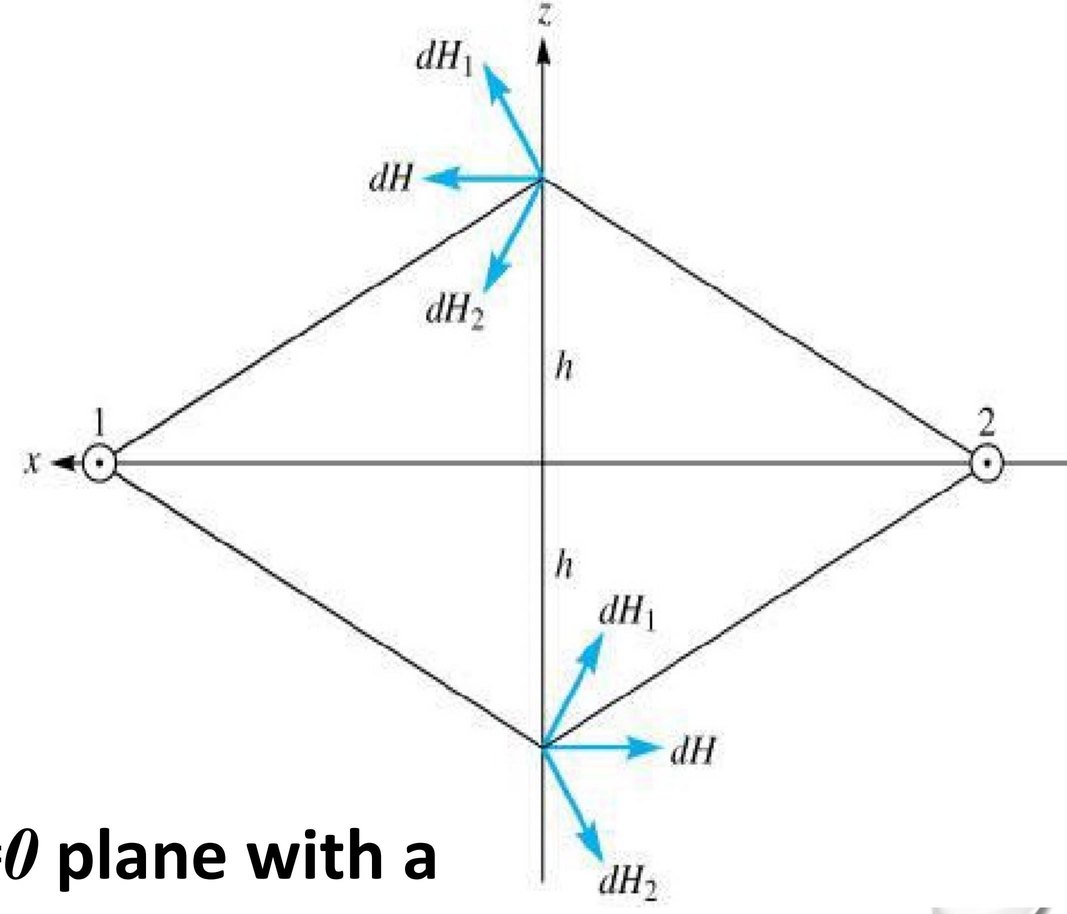
$$\vec{H} = \frac{I}{2\pi\rho}\hat{a}_{\phi}$$



STATIC MAGNETIC FIELD

Ampere’s Circuital law: Infinite Sheet Current





Consider an infinite sheet of current in the $z=0$ **plane with a uniform current density,** $K=K_y\, a_y$.

STATIC MAGNETIC FIELD

Ampere's Circuital law: Infinite Sheet Current

$$I_{enc} = \oint \vec{H} \cdot d\vec{l} = K_y b$$ Apply Ampere's law

$$\vec{H} = \begin{cases} H_0 \hat{a}_x, z > 0 \\ -H_0 \hat{a}_x, z < 0 \end{cases}$$

$$\oint \vec{H} \cdot d\vec{l} = \left(\int_1^2 + \int_2^3 + \int_3^4 + \int_4^1 \right) \vec{H} \cdot d\vec{l}$$

$$= 0(-a) - H_0(-b) + 0(a) + H_0(b) = 2H_0 b$$

STATIC MAGNETIC FIELD

Ampere's Circuital law: Infinite Sheet Current

from Ampere's law and integral summation

$$H_o = K_y$$

$$\vec{H} = \begin{cases} \frac{1}{2} K_y \hat{a}_x, z > 0 \\ -\frac{1}{2} K_y \hat{a}_x, z < 0 \end{cases}$$

Thus for an infinite sheet of charge: $\vec{H} = \frac{1}{2} \vec{K} \times \hat{a}_n$

STATIC MAGNETIC FIELD

Magnetic Flux Density

- **Magnetic Flux density, B, is the magnetic equivalent of the electric flux density, D. As such, one can define:** $\vec{B} = \mu_0 \vec{H}$ where $\mu_0 = 4\pi \times 10^{-7} H/m$

  - **Similarly, from Ampere's Law:** $I_{enc} - \oint \frac{\bar{B}}{\mu_0} \cdot d\hat{l}$

Magnetic flux through a surface is: $\psi = \int_S \vec{B} \cdot d\vec{S} = \mu_0 \int_S \vec{H} \cdot d\vec{S}$

Magnetic flux through an enclosed system: $\psi = \int_S \vec{B} \cdot d\vec{S} = \int_S \left(\nabla \cdot \vec{B}\right) dv$

STATIC MAGNETIC FIELD

Lorentz Force Equation

- The force on a charged particle in an electric field is simply F=qE
- However, in the presence of an electromagnetic field an additional force is imposed from the charge displacement of velocity, u, quantified by the magnetic field, B.
- The combined force is defined by Lorentz Force Law:

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$

Equating the Lorentz force to Newton’s force equation, we have

$$\dot{F} - q(\dot{E} + \bar{u} \times \dot{B}) - m\bar{a} - m\frac{d\bar{u}}{dt}$$

STATIC MAGNETIC FIELD

Force on Current Element

We can use field calculations to determine the force acting on a current element, $Idl = KdS = Jdv$, due to an applied external magnetic field B.

Assume that a copper wire carries a current density, J= ρ_vu.

- **We know:** $I d\vec{l} = \vec{J} dv = \rho_v \vec{u} = dQ\vec{u}$
- **…and that…**

$$\vec{F} = q(\vec{E} + \vec{u} \times \vec{B})$$

$$where \quad \vec{E} \to 0$$

$$\vec{F} = q(\vec{u} \times \vec{B})$$

STATIC MAGNETIC FIELD

Force on Current Element

Thus, we can solve for the force on the first wire:

$$d\vec{F} = dq(\bar{u} \times \vec{B}) = Id\vec{l} \times \vec{B}$$

$$\vec{F} = \oint_L Id\vec{l} \times \vec{B}$$

Likewise

$$\vec{F} = \oint_L \vec{K}dS \times \vec{B} = \vec{F} = \oint_L \vec{J}dv \times \vec{B}$$

STATIC MAGNETIC FIELD

Examples:

A proton moving with a speed of 2×10^6 m/s through a magnetic field with magnetic flux density of 2.5 T experiences a magnetic force of magnitude 4×10^{-13} N. What is the angle between the magnetic field and the proton's velocity?

$$F = quB\sin\theta$$

$$\sin\theta = \frac{F}{quB}$$

$$= \frac{4\times10^{-13}}{1.6\times10^{-19}\times2\times10^{6}\times2.5}$$

$$= 0.5$$

$$\theta = \sin^{-1}0.5 = 30^\circ \text{ or } 150^\circ.$$

STATIC MAGNETIC FIELD

Examples:

A horizontal wire with a mass per unit length of 0.2 kg/m carries a current of 4A in the $+x$-direction. If the wire is placed in a uniform magnetic flux density B, what should the direction and minimum magnitude of B be in order to magnetically lift the wire vertically upward?

(Hint: The acceleration due to gravity is g = −zˆ9.8 m/s².)

Solution: For a length l,

$$\mathbf{F}_g = -\hat{\mathbf{z}}0.2l \times 9.8 = -\hat{\mathbf{z}}1.96l \quad \text{(N)}$$

$$\mathbf{F}_m = \hat{\mathbf{x}}Il \times \mathbf{B}$$

For $F_m + F_g = 0$, F_m has to be along $+\hat{z}$, which means that B has to be along $+\hat{y}$.

STATIC MAGNETIC FIELD

Examples:

A horizontal wire with a mass per unit length of 0.2 kg/m carries a current of 4A in the $+x$-direction. If the wire is placed in a uniform magnetic flux density $\mathbf{B}$, what should the direction and minimum magnitude of $\mathbf{B}$ be in order to magnetically lift the wire vertically upward?

(Hint: The acceleration due to gravity is g = −zˆ9.8 m/s².)

$$1.96l = IlB$$

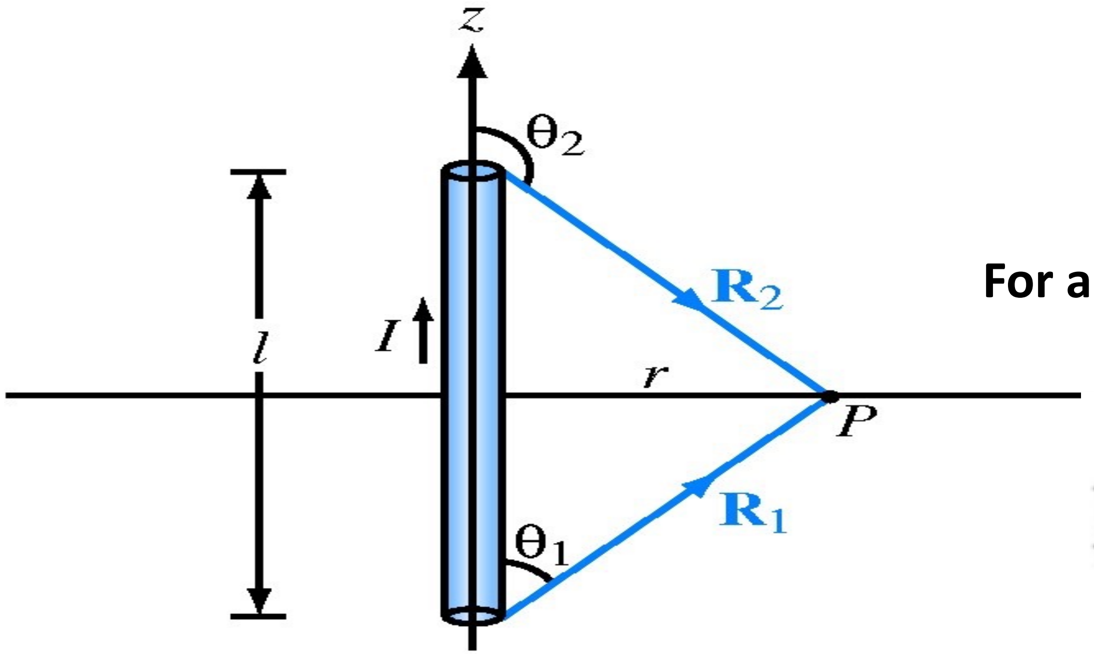
$$B = \frac{1.96}{I} = 0.49 \text{ (T), and}$$

$$\mathbf{B} = \hat{\mathbf{y}}\,0.49 \text{ (T)}.$$

STATIC MAGNETIC FIELD

Examples:

A semi-infinite linear conductor extends between $z = 0$ and $z = \infty$ along the z-axis. If the current I in the conductor flows along the positive z-direction, find H at a point in the x–y plane at a radial distance r from the conductor.



$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r}(\cos\theta_1 - \cos\theta_2)$$

For a conductor extending from $z = 0$ to $z = \infty$, $\theta_1 = 0$ and $\theta_2 = \pi$. Hence,

$$\mathbf{H} = \hat{\boldsymbol{\phi}} \frac{I}{4\pi r}(1+1) = \hat{\boldsymbol{\phi}} \frac{I}{2\pi r} \quad \text{(A/m)}.$$

TOPIC-3 AREAS:

TIME VARYING ELECTROMAGNETIC FIELD:

- **Inductors and Inductance**
- **Magnetic Energy**
- **Faraday’s Law for induced emf:**
- **Maxwell’s Time Dependent Equations**
- **Magnetic Flux Density etc.**

TIME VARYING EM'TIC FIELDS

Inductors and Inductance:

For magnetic circuit carrying current I producing a magnetic field with flux…

$$\Psi = \int \bar{B} \cdot d\bar{S}$$

We define the flux linkage between a circuit with N identical turns as,

$$\lambda = N\Psi$$

As flux linkage is proportional to the current I producing it thus, written as

$$\lambda = LI$$

Where L = circuit inductance

Measured in Henrys (H) = Wb/A.

TIME VARYING EM'TIC FIELDS

Inductors and Inductance:

We can equate the inductance to the magnetic flux of the circuit as…

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I}$$

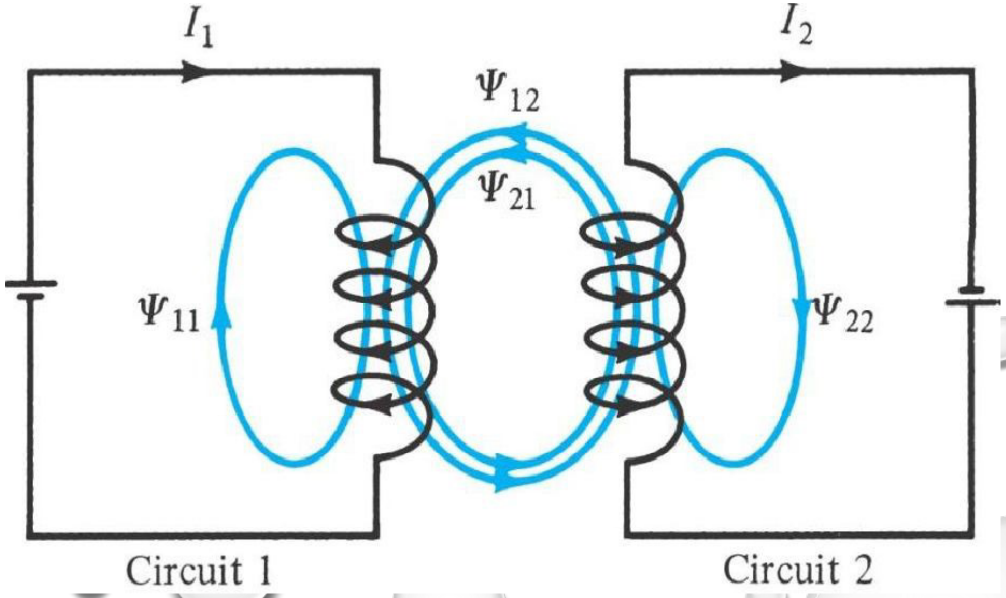
The magnetic energy (in Joules) stored by the inductor is expressed as,

$$W_m = \frac{1}{2} L I^2$$

TIME VARYING EM'TIC FIELDS

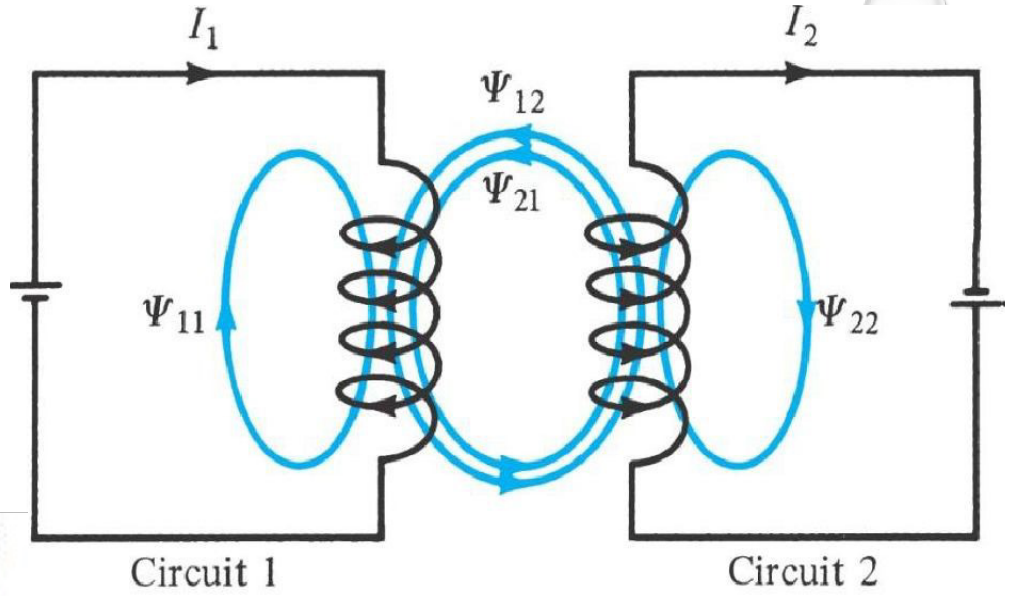
Inductors and Inductance:

Since magnetic fields produce forces on nearby current elements, and magnetic fields can be generated by an isolated or coupled set of current carrying circuits, then it is only reasonable that such circuits may induce fields and magnetization between them…



TIME VARYING EM'TIC FIELDS

Inductors and Inductance:



We can calculate the individual flux linkage between the two components as …

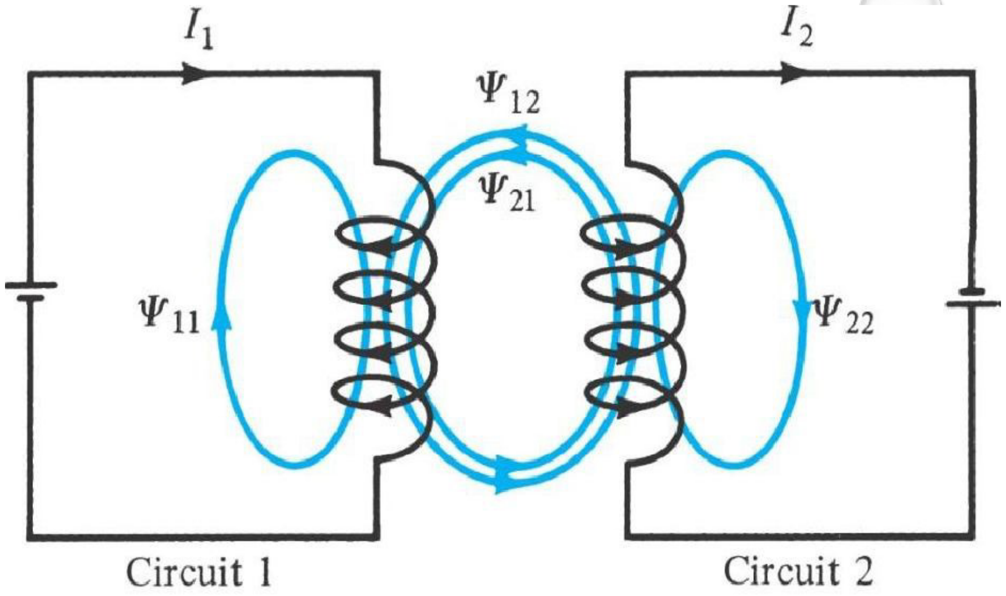
$$\Psi_{12} = \int_{S_1} \vec{B}_2 \cdot d\vec{S}$$

Also, the mutual inductance between the circuits that is equal from circuit 12 as it is from circuit 21 as

$$M_{12} = \frac{\lambda_{12}}{I_2} = \frac{N_1 \Psi_{12}}{I_2}$$

TIME VARYING EM'TIC FIELDS

Inductors and Inductance:



Individual inductances are…

$$L_1 = \frac{\lambda_{11}}{I_1} = \frac{N_1 \Psi_1}{I_1} \qquad L_2 = \frac{\lambda_{22}}{I_2} = \frac{N_2 \Psi_2}{I_2}$$

The total magnetic energy in the circuit is…

$$W_m = W_1 + W_2 + W_{12} = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 \pm M_{12} I_1 I_2$$

TIME VARYING EM'TIC FIELDS

Magnetic Energy:

For magnetic energy as a function of inductance…

$$W_m = \frac{1}{2} L I^2$$

$$\Delta L = \frac{\Delta \Psi}{\Delta I} = \frac{\mu H \Delta x \Delta y}{\Delta I}$$

$$where, \Delta I = H \Delta z$$

$$\Delta L = \frac{\mu H \Delta x \Delta y}{\Delta I} = \frac{\mu \Delta x \Delta y}{\Delta z}$$

$$\Delta W_m = \frac{1}{2} \Delta L \Delta I^2 = \frac{1}{2} \mu H^2 \Delta x \Delta y \Delta z$$

$$\Delta W_m = \frac{1}{2} \mu H^2 \Delta v$$

$$w_m = \lim_{\Delta v \to 0} \frac{\Delta W_m}{\Delta v} = \frac{1}{2} \mu H^2 = \frac{1}{2} \left(\vec{H} \cdot \vec{B} \right) = \frac{B^2}{2\mu}$$

$$W_m = \int w_m dv = \int \frac{1}{2} \left(\vec{H} \cdot \vec{B} \right) dv = \int \frac{1}{2} \mu H^2 dv$$

TIME VARYING EM'TIC FIELDS

Magnetic Circuit:

The following relations allow one to solve magnetic field problems…

Electric	Magnetic
Conductivity σ	Permeability μ
Field intensity E	Field intensity H
Current $I = \int \mathbf{J} \cdot d\mathbf{S}$	Magnetic flux $\Psi = \int \mathbf{B} \cdot d\mathbf{S}$
Current density $J = \dfrac{I}{S} = \sigma E$	Flux density $B = \dfrac{\Psi}{S} = \mu H$
Electromotive force (emf) V	Magnetomotive force (mmf) $\mathcal{F}$
Resistance R	Reluctance $\mathcal{R}$
Conductance $G = \dfrac{1}{R}$	Permeance $\mathcal{P} = \dfrac{1}{\mathcal{R}}$
Ohm's law $R = \dfrac{V}{I} = \dfrac{\ell}{\sigma S}$	Ohm's law $\mathcal{R} = \dfrac{\mathcal{F}}{\Psi} = \dfrac{\ell}{\mu S}$
or $V = E\ell = IR$	or $\mathcal{F} = H\ell = \Psi\mathcal{R} = NI$
Kirchhoff's laws:	Kirchhoff's laws:
$\Sigma I = 0$	$\Sigma \Psi = 0$
$\Sigma V - \Sigma RI = 0$	$\Sigma \mathcal{F} - \Sigma \mathcal{R} \Psi = 0$

TIME VARYING EM'TIC FIELDS

Faradays LAW (of induced EMF):

Faraday's law state that… the Induced electromotive force (emf) (in volts) in any closed circuit is equal to the time rate of change of magnetic flux by the circuit.

$$V_{emf} = -\frac{d\lambda}{dt} = -N\frac{d\Psi}{dt}$$

Where: λ = is the flux linkage,

ψ = is the magnetic flux,

N = is the number of turns in the inductor, and

t = represents a time interval.

Negative (-) sign shows that the induced voltage acts to oppose the flux producing it.

TIME VARYING EM'TIC FIELDS

Transformer and Motional Electromotive Forces:

The variation of flux with time may be caused by three ways

1. **Having a stationary loop in a time-varying B field**

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$

2. **Having a time-varying loop in a static B field**

$$\nabla \times \vec{E}_m = \nabla \times \left(\vec{u} \times \vec{B}\right)$$

3. **Having a time-varying loop in a time-varying B field**

$$\nabla \times \vec{E}_m = -\frac{d\vec{B}}{dt} + \nabla \times \left(\vec{u} \times \vec{B}\right)$$

TIME VARYING EM'TIC FIELDS

Displacement Current:

Lets now examine time dependent fields from Ampere's Law .

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \cdot \left(\nabla \times \vec{H}\right) = 0 = \nabla \cdot \vec{J}$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t} \neq 0$$

This vector identity for the cross product is mathematically valid. However, it requires that the continuity eqn. equals zero, which is not valid from an electrostatics standpoint!

$$\nabla \times \vec{H} = \vec{J} + \vec{J}_d$$

Thus, lets add an additional current density term to balance the electrostatic field requirement

$$\nabla \cdot \left(\nabla \times \vec{H}\right) = 0 = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}_d$$

TIME VARYING EM'TIC FIELDS

Displacement Current:

Lets now examine time dependent fields from Ampere's Law .

$$\nabla \cdot \vec{J}_d = -\nabla \cdot \vec{J} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t}\left(\nabla \cdot \vec{D}\right) = \nabla \cdot \frac{\partial D}{\partial t}$$

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

We can now define the displacement current density as the time derivative of the displacement vector

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

Another of Maxwell's for time varying fields

This one relates Magnetic Field Intensity to conduction and displacement current densities

TIME VARYING EM'TIC FIELDS

Displacement Current:

Lets now examine time dependent fields from Ampere's Law .

$$\nabla \cdot \vec{J}_d = -\nabla \cdot \vec{J} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t}\left(\nabla \cdot \vec{D}\right) = \nabla \cdot \frac{\partial D}{\partial t}$$

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

We can now define the displacement current density as the time derivative of the displacement vector

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

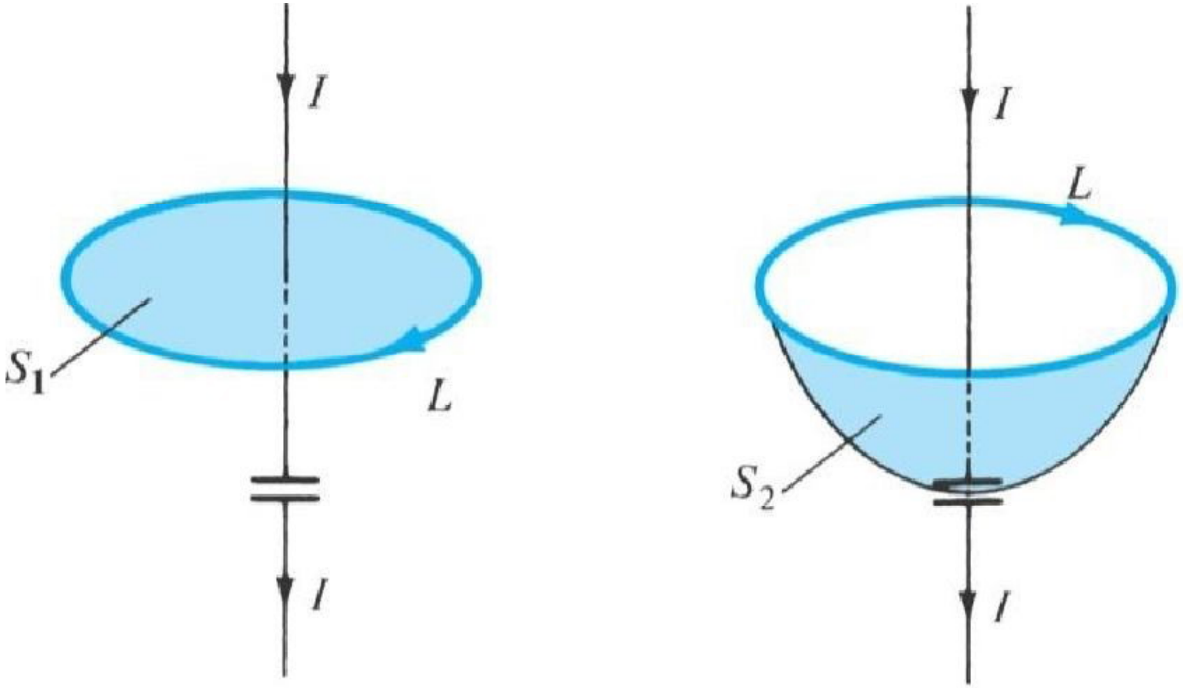
Another of Maxwell's for time varying fields

This one relates Magnetic Field Intensity to conduction and displacement current densities

TIME VARYING EM'TIC FIELDS

Displacement Current:

We can apply the displacement current concept on the simple case of a capacitive element in a simple electronic circuit, as shown below.



TIME VARYING EM'TIC FIELDS

Displacement Current:

Applying Ampere's circuit law to a closed path provides the following eqn. for current on the first side of the capacitive element.

surface 2 is the opposite side of the capacitor and has no conduction current allowing for no enclosed current at surface 2. If $J = 0$ on the second surface, then Jd must be generated on the second surface to create a time displaced current equal to current on surface 1.

TIME VARYING EM'TIC FIELDS

Displacement Current:

$$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$$

$$I_d = \int \bar{J} \cdot d\bar{S} = \int \frac{\partial \bar{D}}{\partial t} \cdot d\bar{S}$$

$$\oint_L \bar{H} \cdot d\bar{l} = \int_{S_1} \bar{J} \cdot d\bar{S} = I_{enc} = I$$

$$\oint_L \bar{H} \cdot d\bar{l} = \int_{S_2} \bar{J} \cdot d\bar{S} = I_{enc} = 0$$

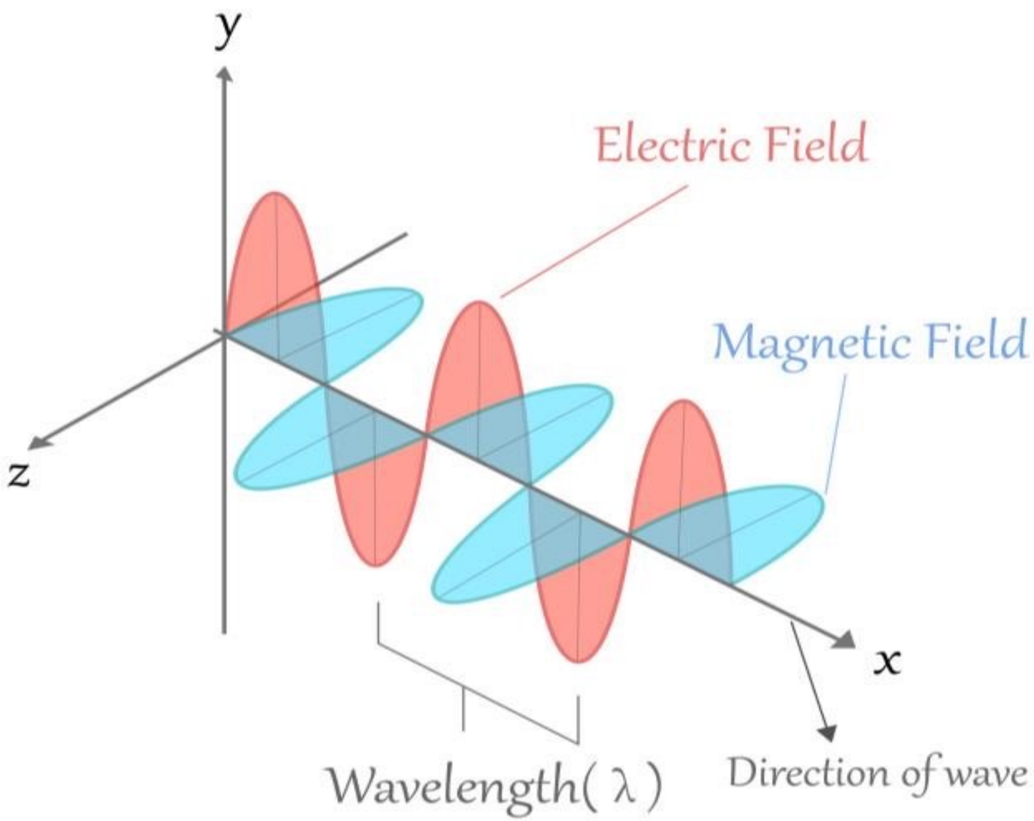
TOPIC-4 AREAS:

ELECTROMAGNETIC WAVE:

- **Plane Wave**
- **Wave Equation.**

INTRODUCTION

ELECTROMAGNETIC WAVE:



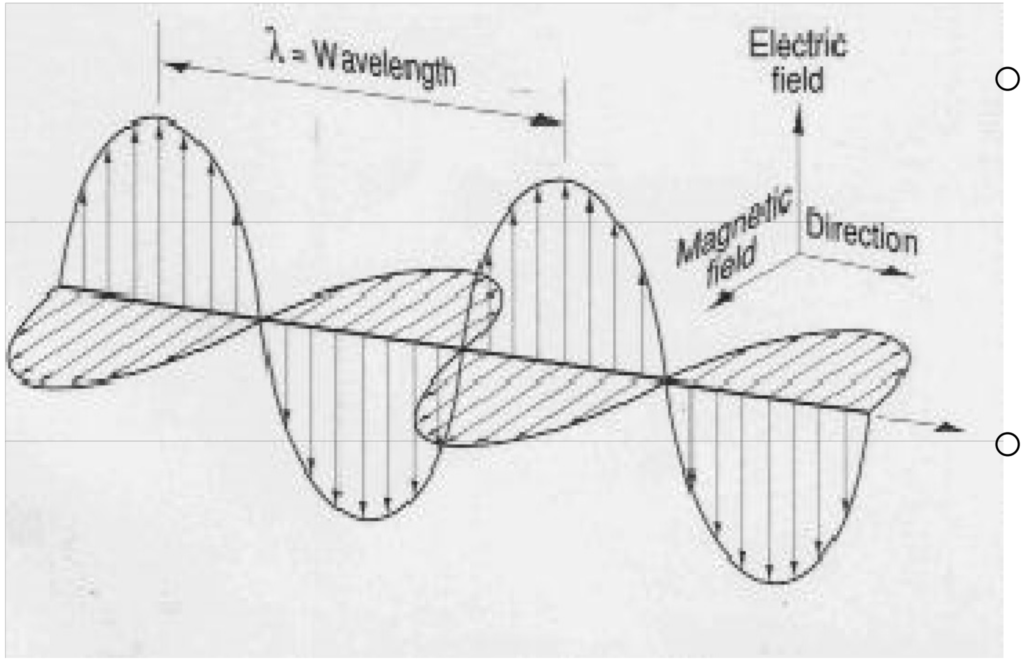
- Generally, plane wave has dependence thus, propagation in a particular direction.

- Usually described using 3-dimentional axes x, y, z components.

- From Electromagnetic Field Theory studied in the previous sessions, several laws, theorem, gives mathematical background:
  - ✓ Faradays law
  - ✓ Biot Savat Law
  - ✓ Maxwells's Equation etc.

INTRODUCTION

ELECTROMAGNETIC WAVE:



- **wavelength** λ to be the distance between successive peaks.

- As shown, the propagating in the z direction, thus, wave has no dependence in the x or y direction.

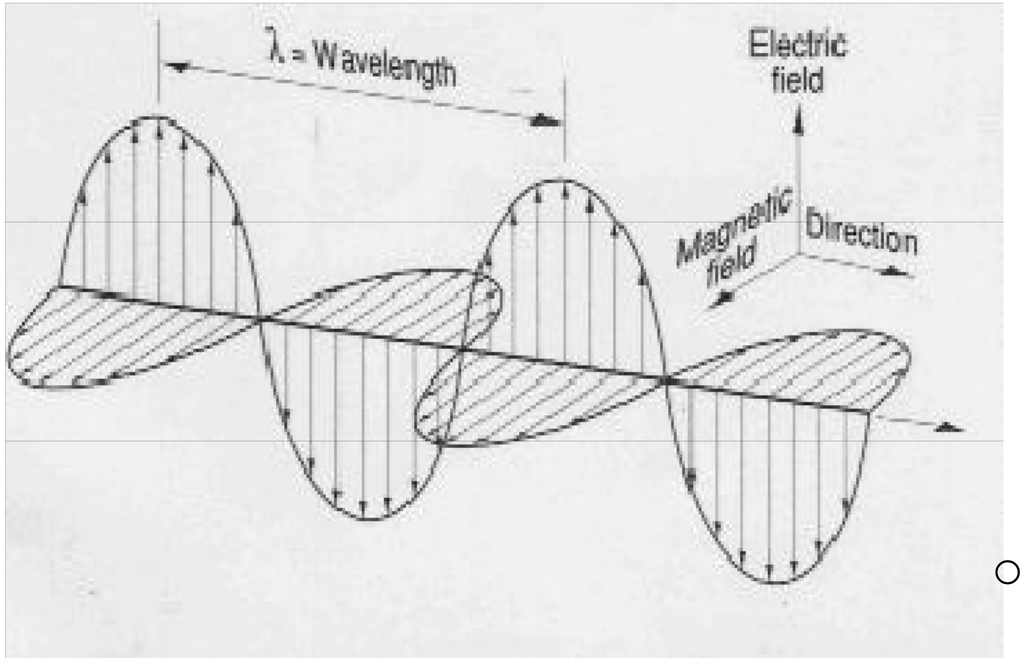
$$E_y(x, y, z, t) = E_y(z, t) \qquad \text{Eqn. 1}$$

- If the wave time harmonic is known, the z and t dependence can be separated as follows:

$$E_y(z, t) = E_y(z)e^{j\omega t} \qquad \text{Eqn. 2}$$

INTRODUCTION

ELECTROMAGNETIC WAVE:



Investigating how Eqn. 2 will satisfy the wave equation.

- First, we'll determine the second derivative of E_y with respect to z:

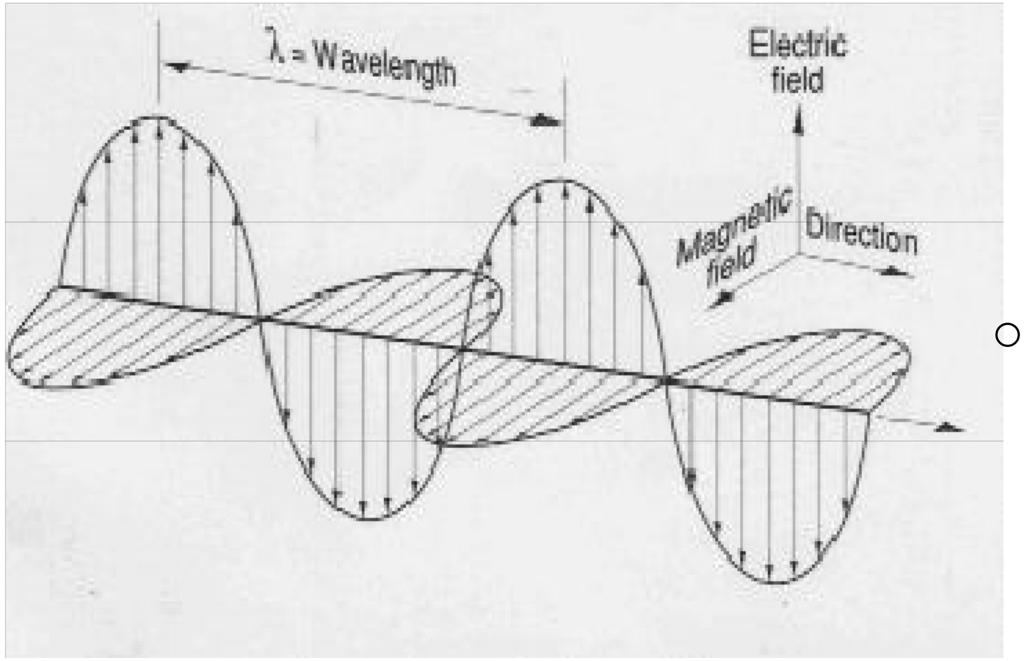
$$\frac{\partial^2 E_y(z,t)}{\partial z^2} = \frac{\partial^2 E_y(z)}{\partial z^2} e^{j\omega t} \quad \text{Eqn. 3}$$

- Next, we'll calculate the second derivative with respect to t:

$$\frac{\partial^2 E_y(z,t)}{\partial t^2} = E_y(z)(j\omega)^2 e^{j\omega t} \quad \text{Eqn. 4}$$

INTRODUCTION

ELECTROMAGNETIC WAVE:



- Substituting Equations 3 and 4 into the wave equation 2 above, we find:

$$\frac{\partial^2 E_y(z)}{\partial z^2} e^{j\omega t} - \frac{1}{c^2} E_y(z)(j\omega)^2 e^{j\omega t} = 0 \qquad \text{Eqn. 5}$$

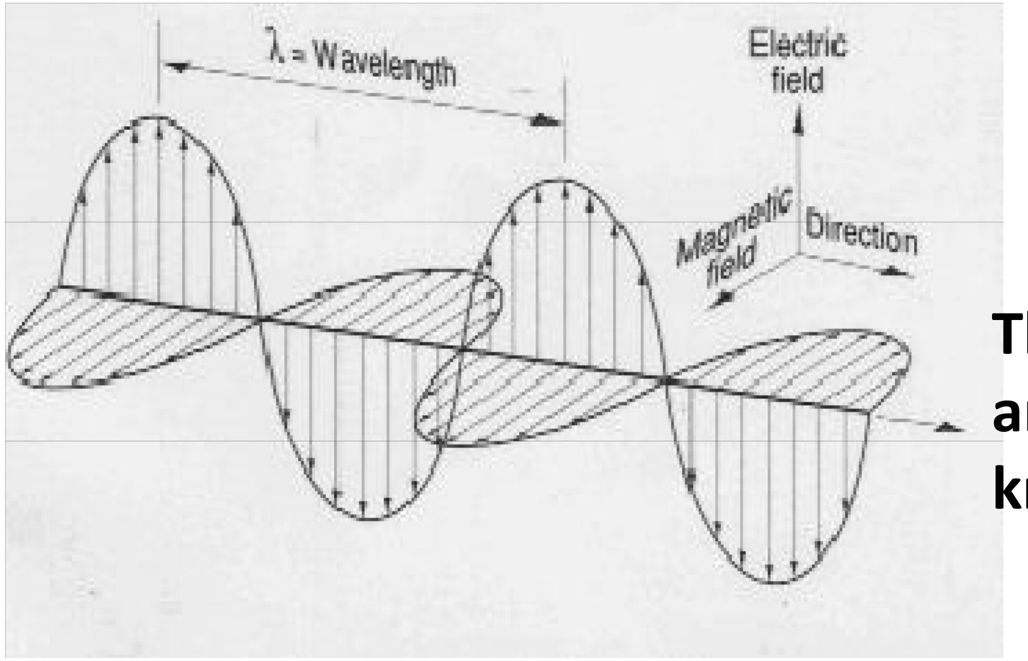
- This can be simplified as follows

$$\frac{\partial^2 E_y(z)}{\partial z^2} + \left(\frac{\omega}{c}\right)^2 E_y(z) = 0 \qquad \text{Eqn. 6}$$

- If k (called the wave number) is define as: $k = \frac{\omega}{c}$

INTRODUCTION

ELECTROMAGNETIC WAVE:



- Then Equation 6 simplifies:

$$\frac{\partial^2 E_y(z)}{\partial z^2} + k^2 E_y(z) = 0$$ Eqn. 8

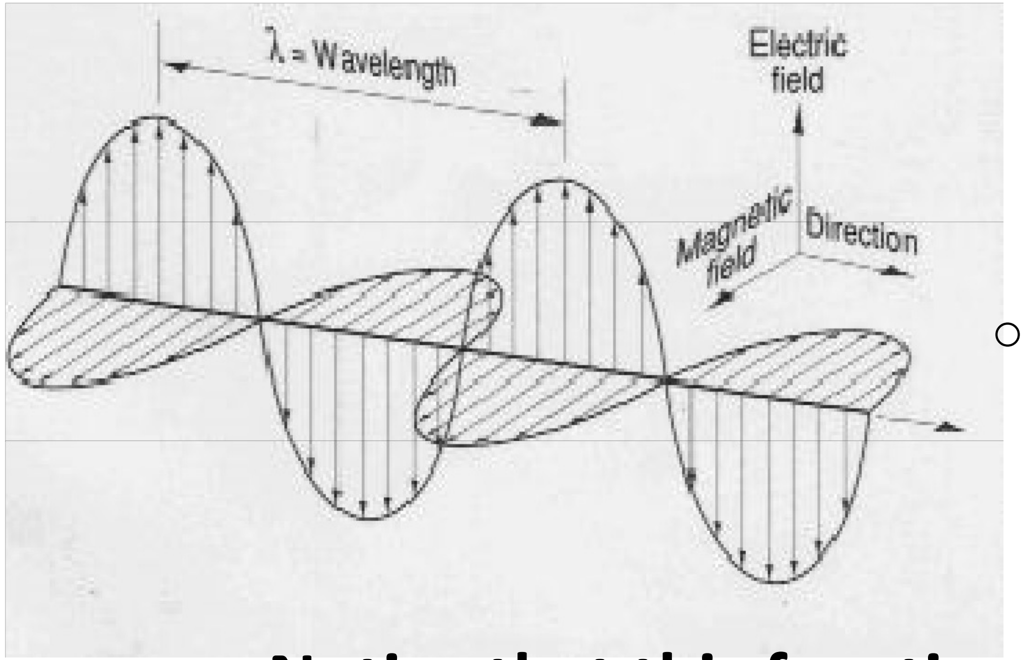
This is known as one-dimensional *Helmholtz* equation, and it is a well-known differential equation with a well-known solution:

$$E_y(z) = ae^{-jkz} + be^{jkz}$$ Eqn. 9

- Adding the time dependence back into this result, we obtain:

INTRODUCTION

ELECTROMAGNETIC WAVE:



- Adding the time dependence back into this result, we obtain:

$$E_y(z,t) = ae^{-jkz}e^{j\omega t} + be^{jkz}e^{j\omega t}$$ Eqn. 10

- Which can be written as:

$$E_y(z,t) = ae^{j(\omega t - kz)} + be^{j(\omega t + kz)}$$ Eqn. 11

Notice that this function has the form of Eqn. 2 as long as Eqn. 1 is followed, so it is guaranteed to be a solution to the wave equation.

INTRODUCTION

ELECTROMAGNETIC WAVE:

- **In order to ensure that Equation 11 is a solution to the wave equation, the following relationship must be true.**

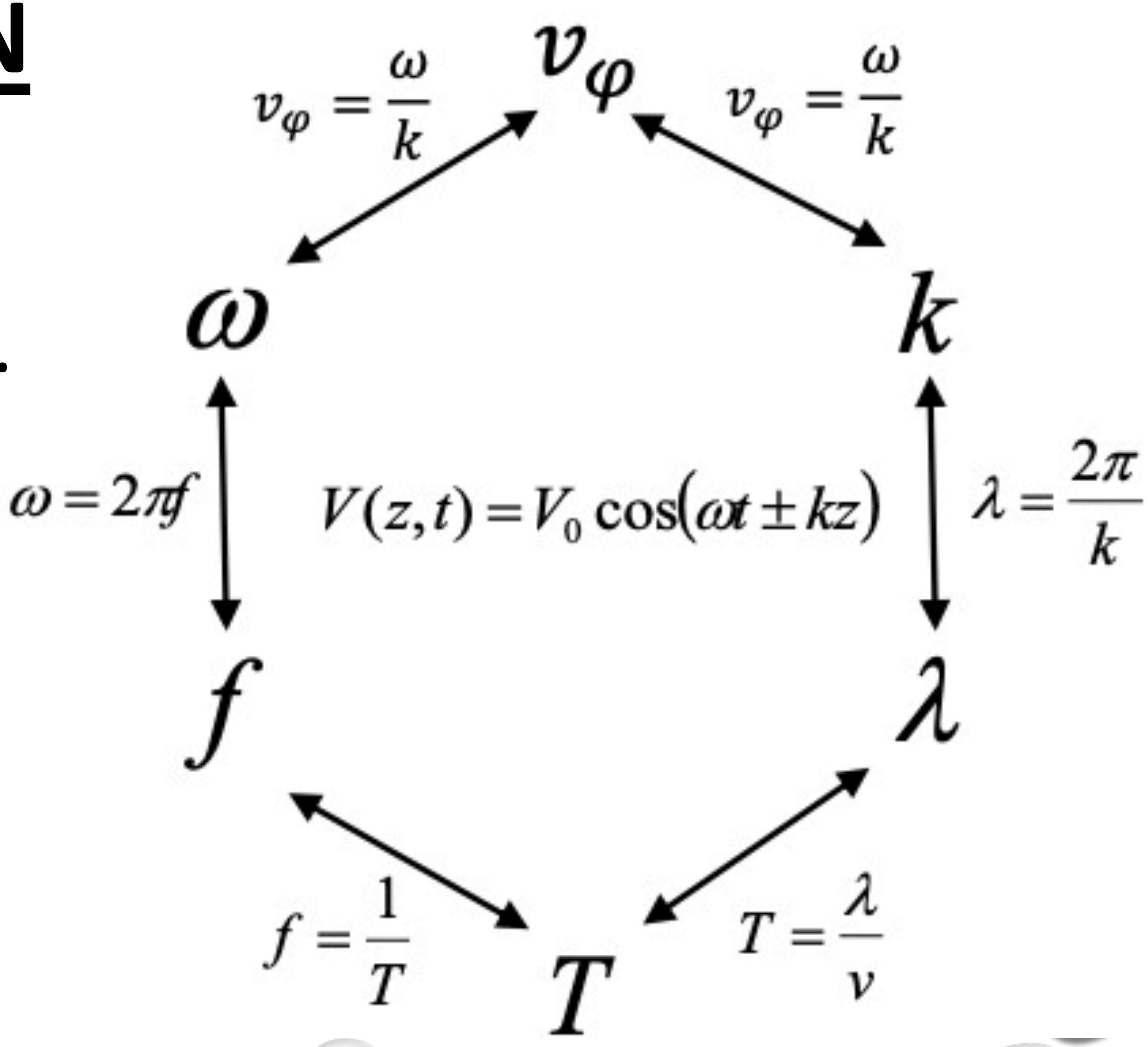
$$v_\phi = \frac{dz}{dt} = \pm c = \pm\frac{\omega}{k}$$

- **$\pm$ signs correspond to a right - or - left-moving wave. NOTE the variables that describe the time and spatial variation of $E(z, t)$:**
  - ✓ **Wavelength λ**
  - ✓ **Wave number k,**
  - ✓ **Radial frequency ω,**
  - ✓ **Frequency f,**
  - ✓ **Period T,**
  - ✓ **Phase velocity $v\phi$. .**

<u>INTRODUCTION</u>

ELECTROMAGNETIC WAVE:

To graphically demonstrate the relationships among these variables.



If you know two of the variables, you will be able to calculate the other four.